

BFS Traversal

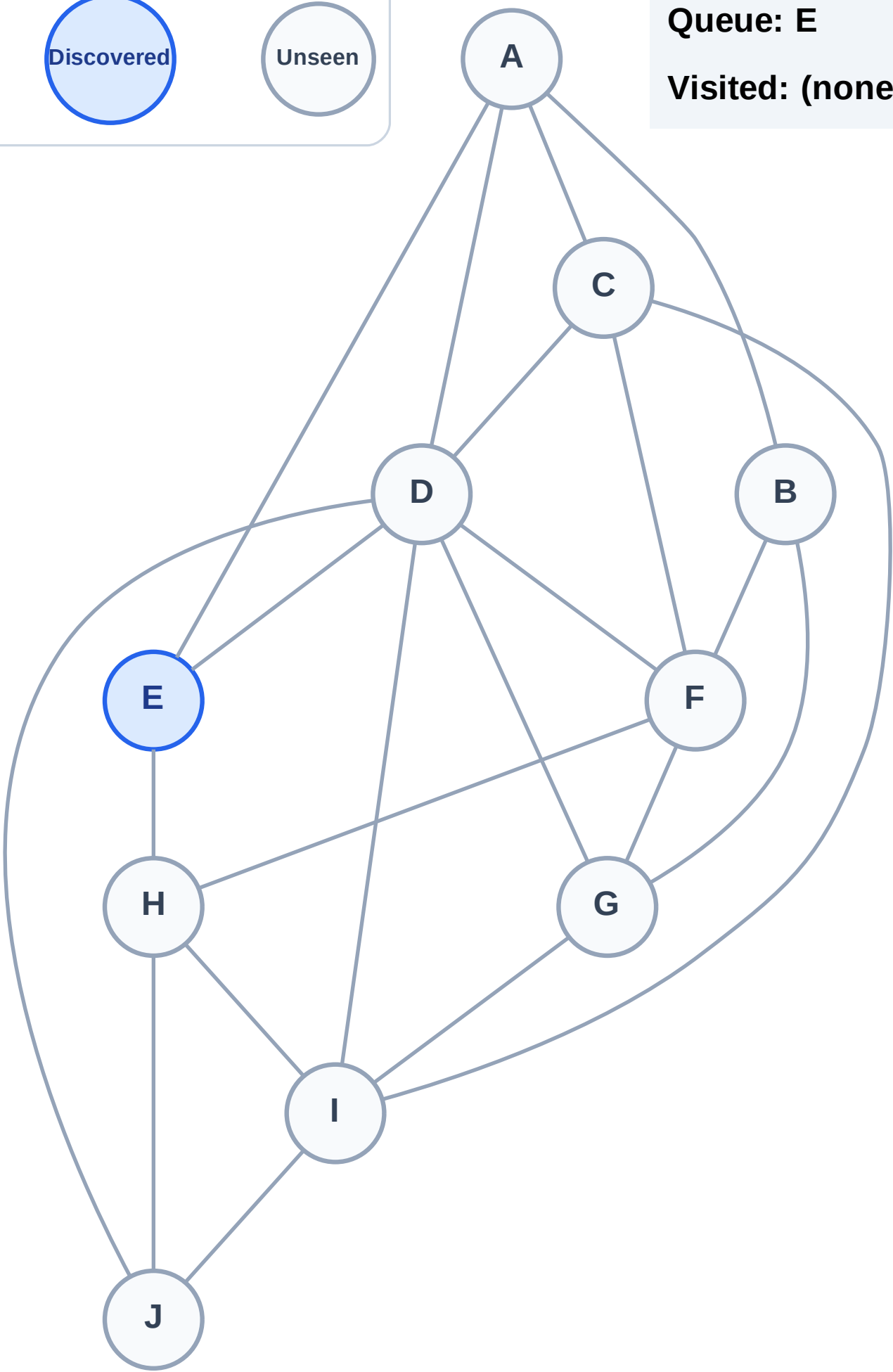
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

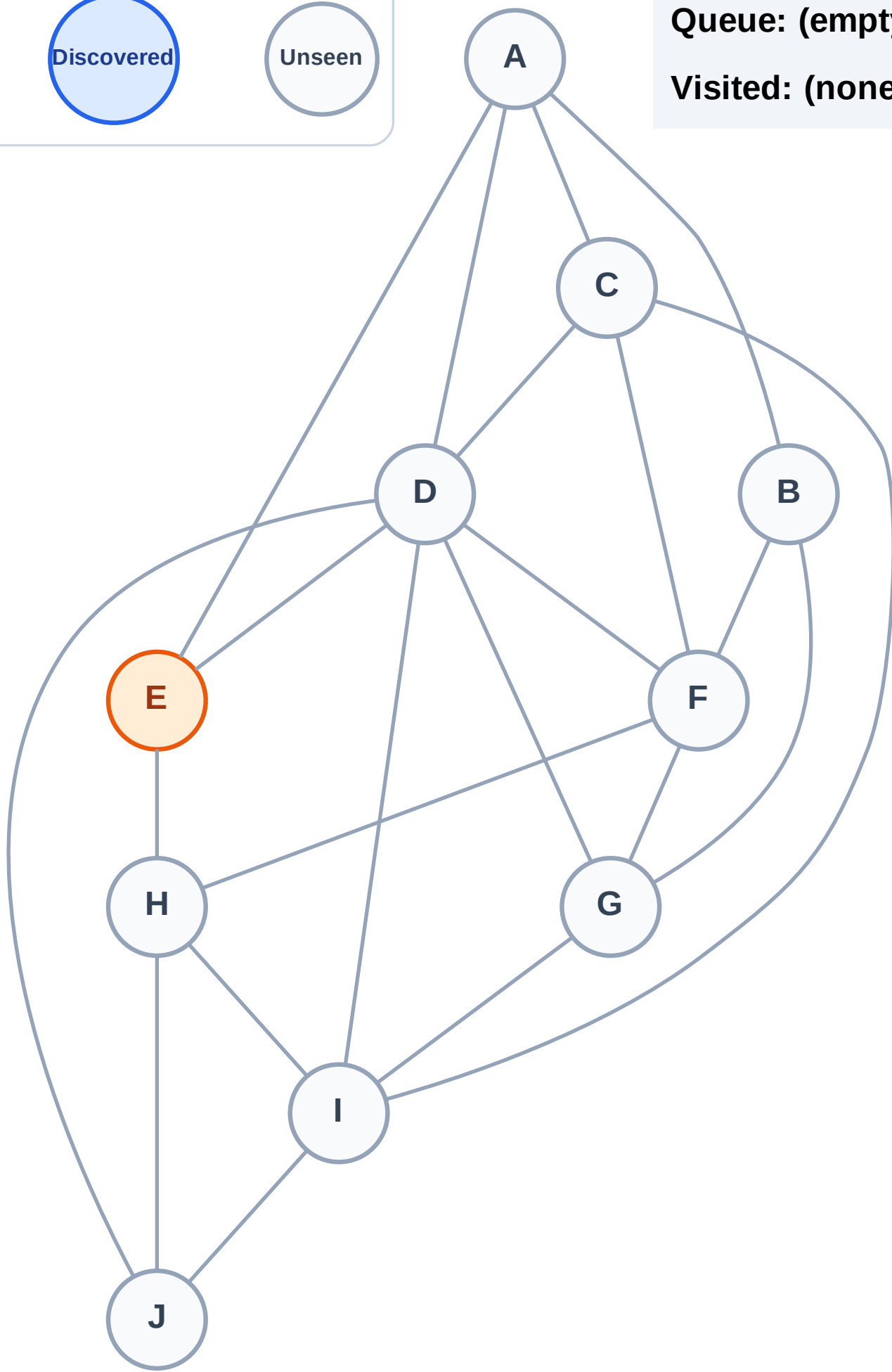
Step 2: Process node E

Legend



Queue: (empty)

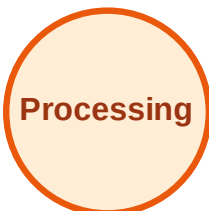
Visited: (none)



BFS Traversal

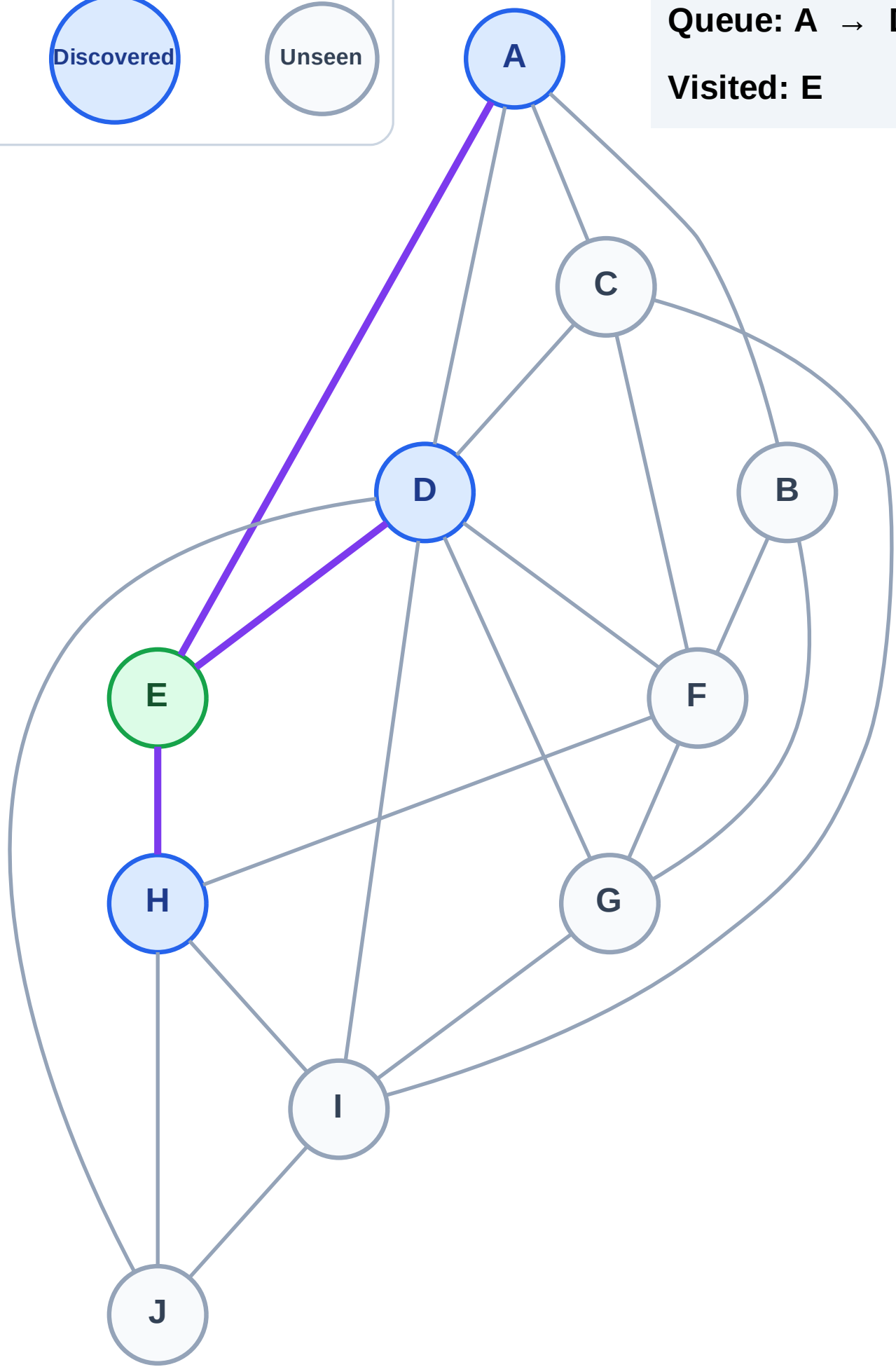
Step 3: Discover A, D, H from E

Legend



Queue: A → D → H

Visited: E



BFS Traversal

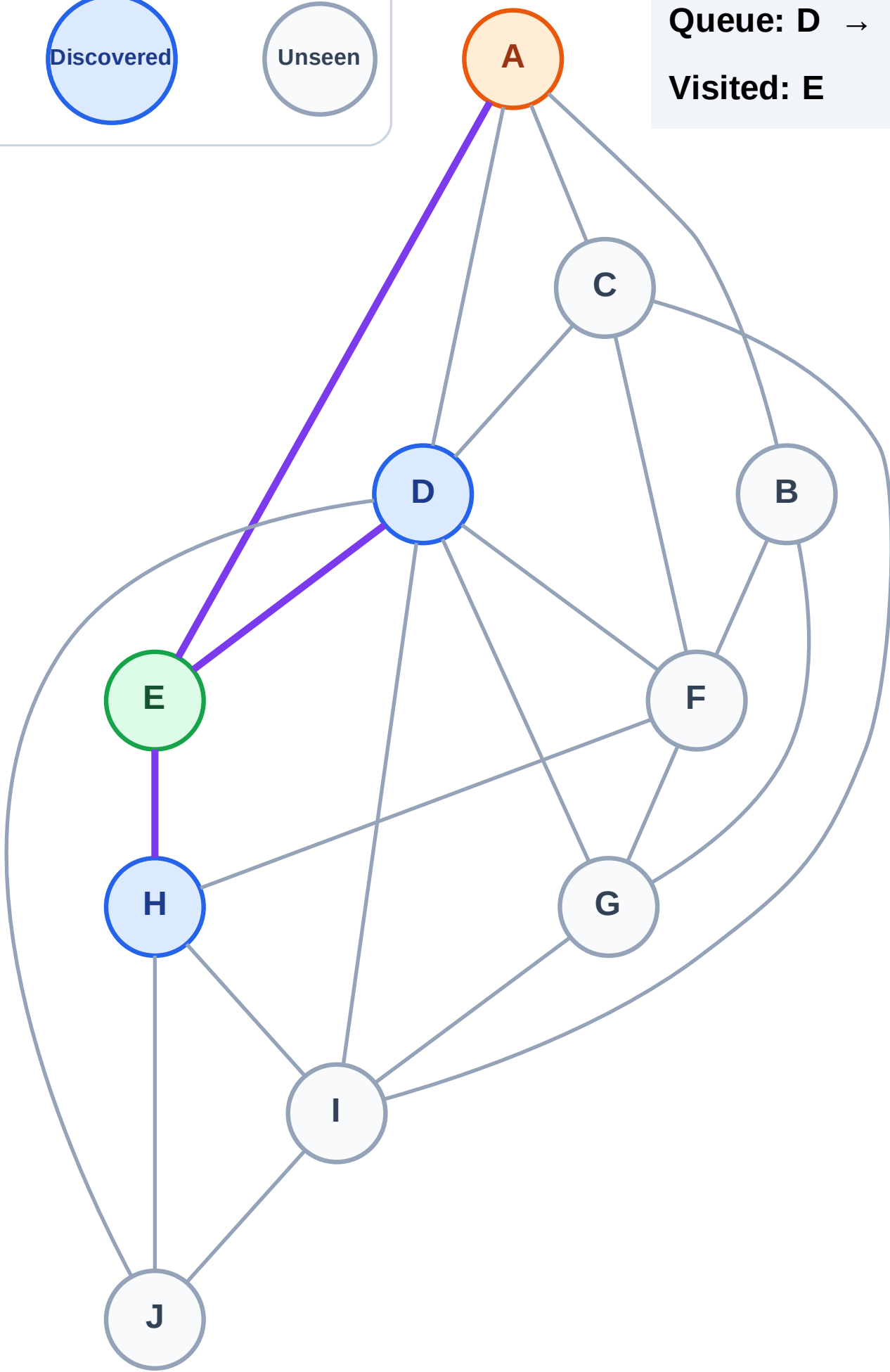
Step 4: Process node A

Legend



Queue: D → H

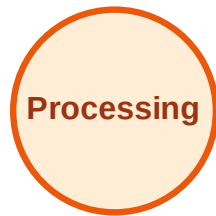
Visited: E



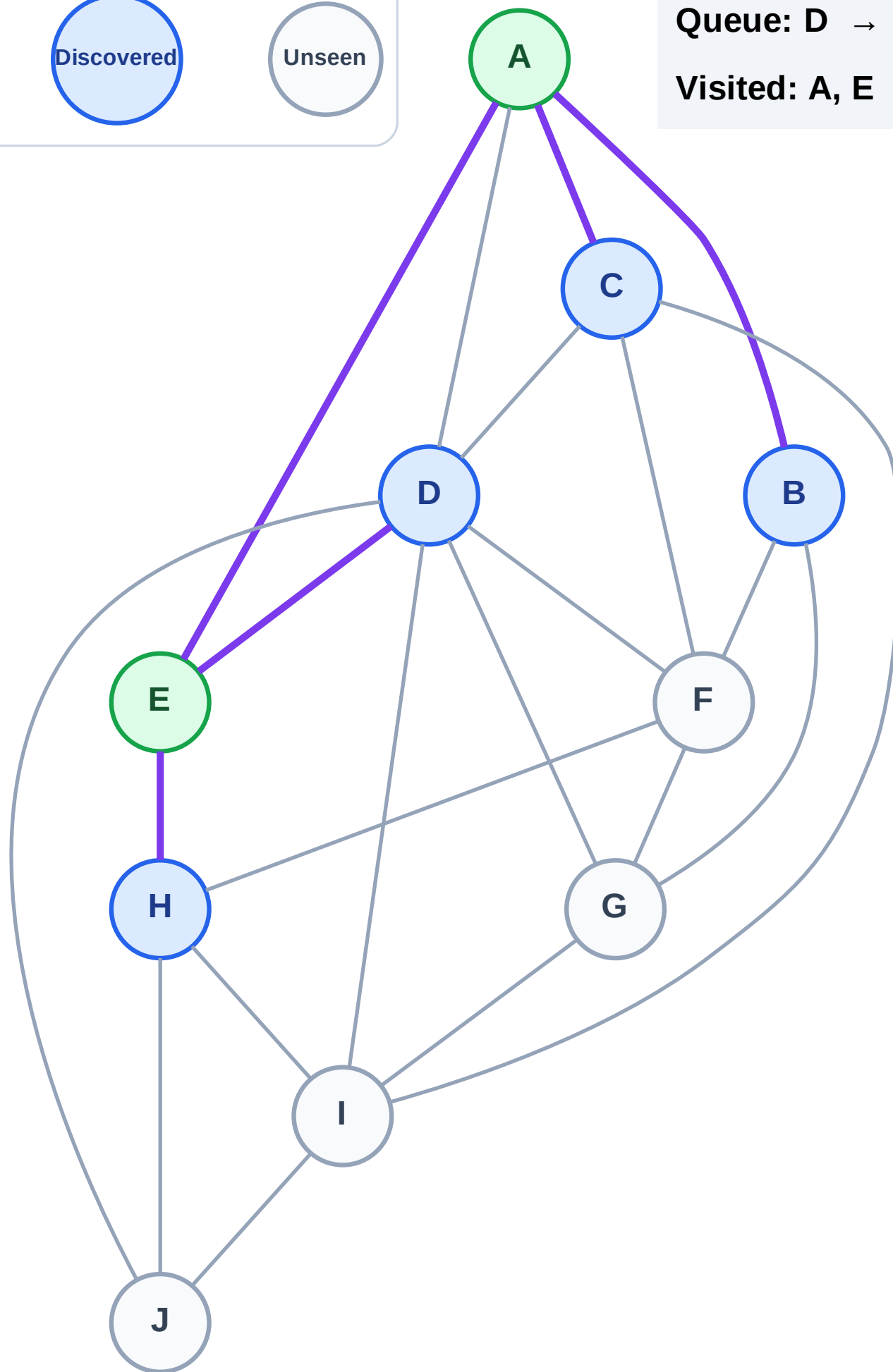
BFS Traversal

Step 5: Discover B, C from A

Legend



Queue: D → H → B → C
Visited: A, E



BFS Traversal

Step 6: Process node D

Legend

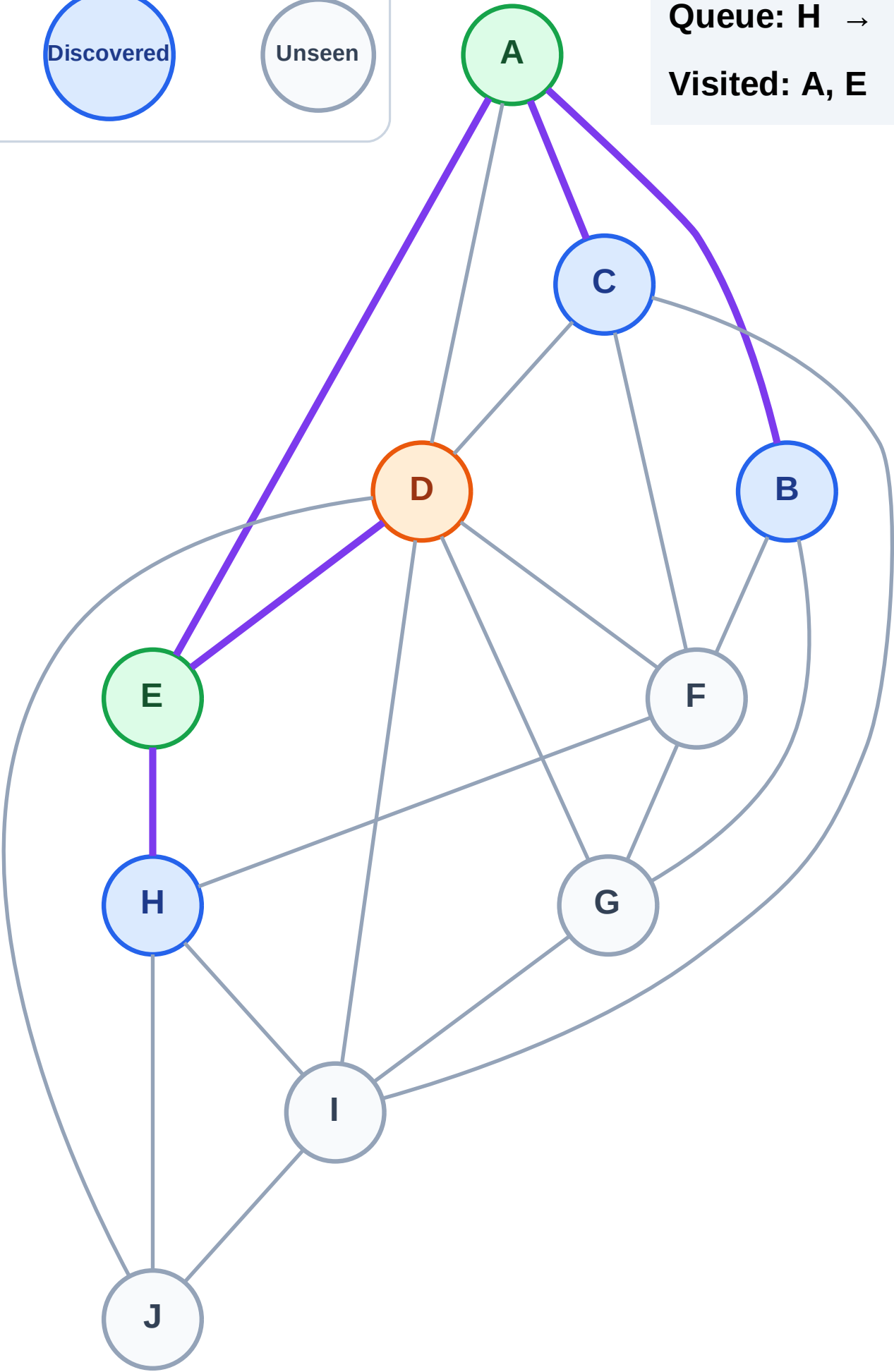
Complete

Processing

Discovered

Unseen

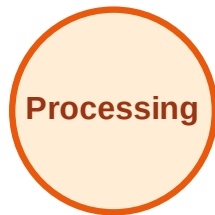
Queue: H → B → C
Visited: A, E



BFS Traversal

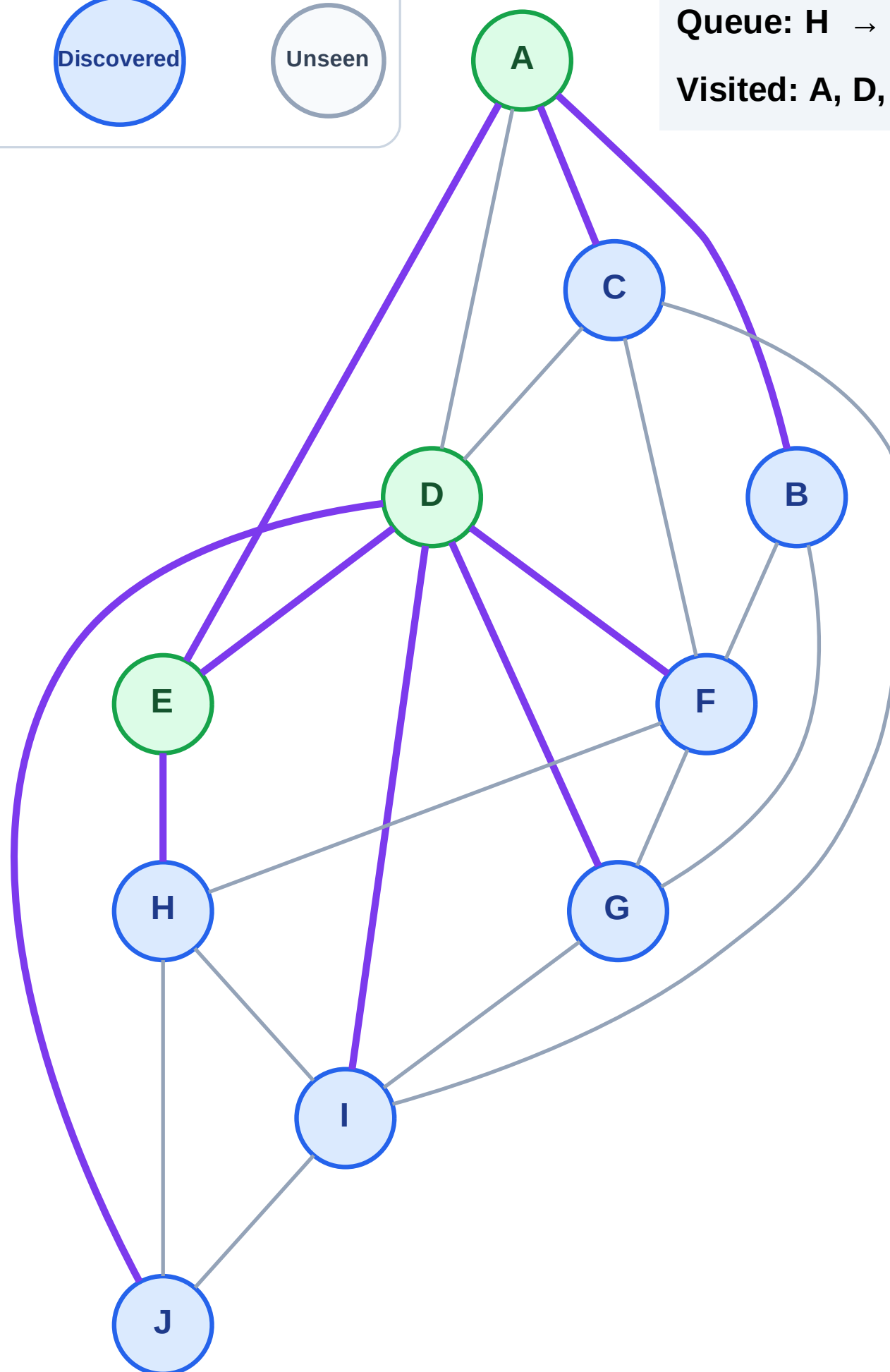
Step 7: Discover F, G, I, J from D

Legend



Queue: H → B → C → F → G → I → J

Visited: A, D, E



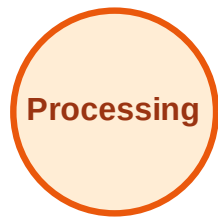
BFS Traversal

Step 8: Process node H

Legend



Complete



Processing



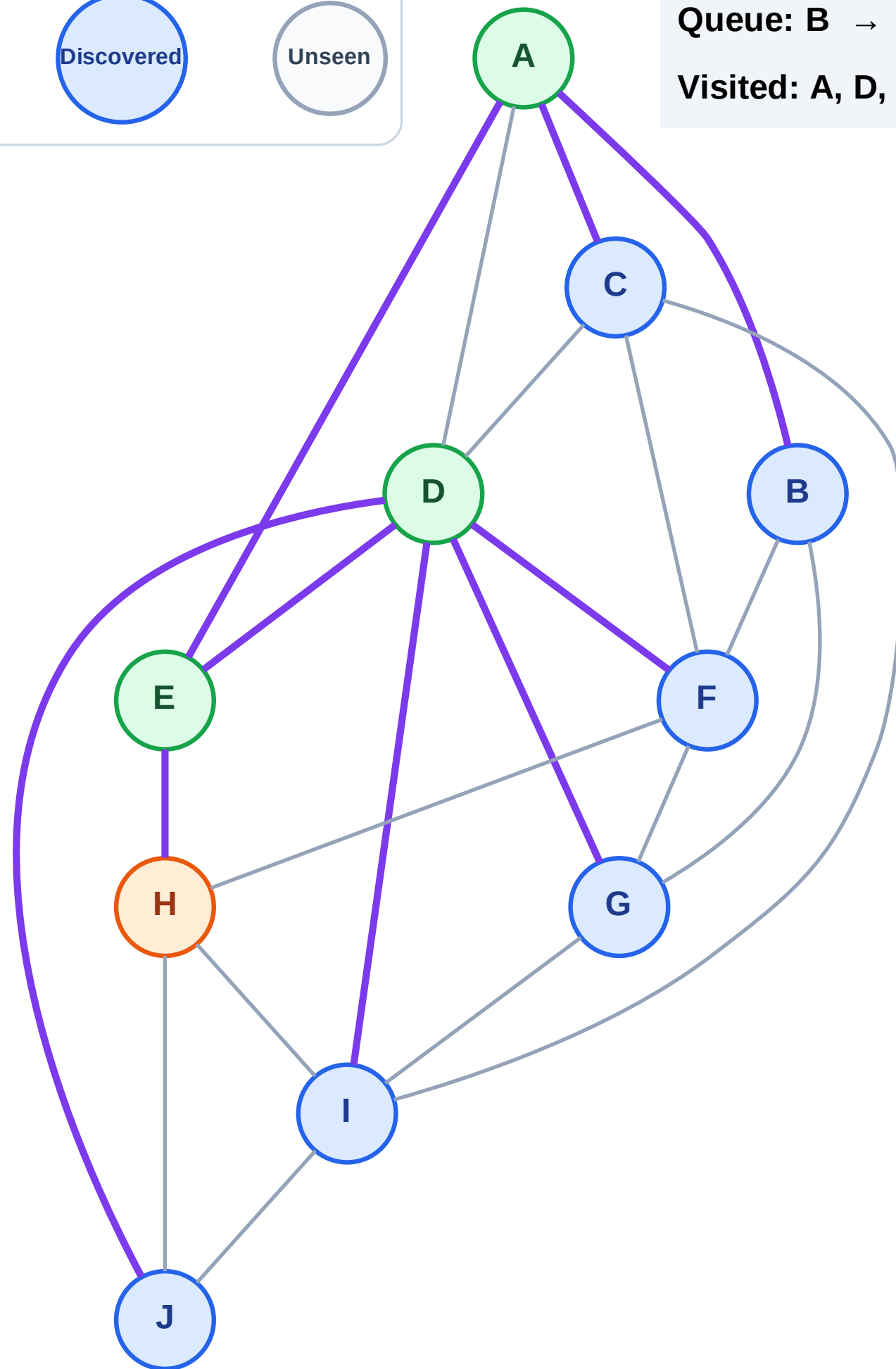
Discovered



Unseen

Queue: B → C → F → G → I → J

Visited: A, D, E



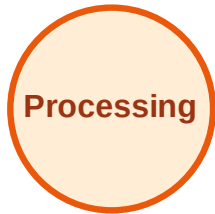
BFS Traversal

Step 9: No new neighbors from H

Legend



Complete



Processing



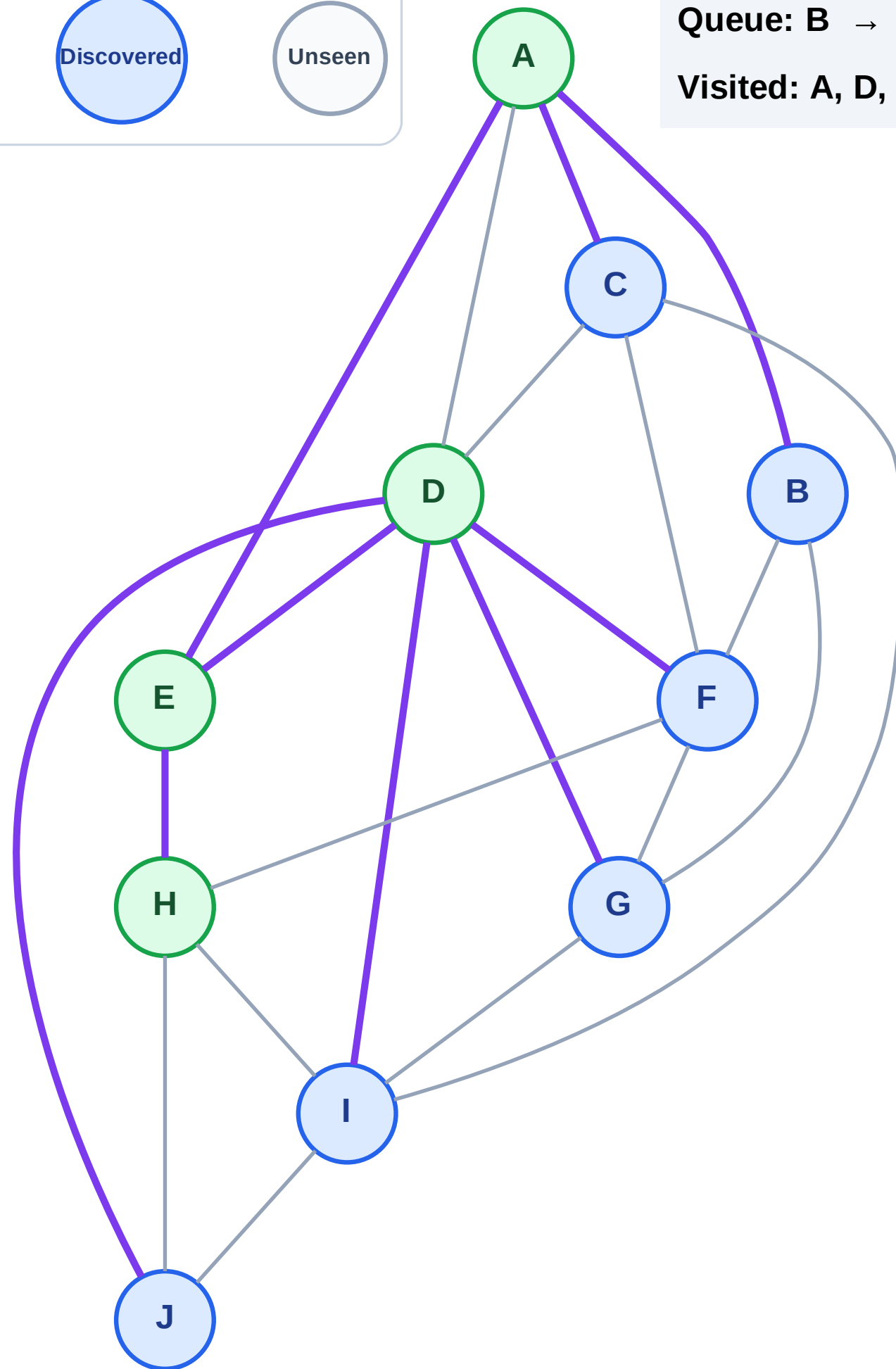
Discovered



Unseen

Queue: B → C → F → G → I → J

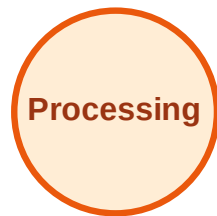
Visited: A, D, E, H



BFS Traversal

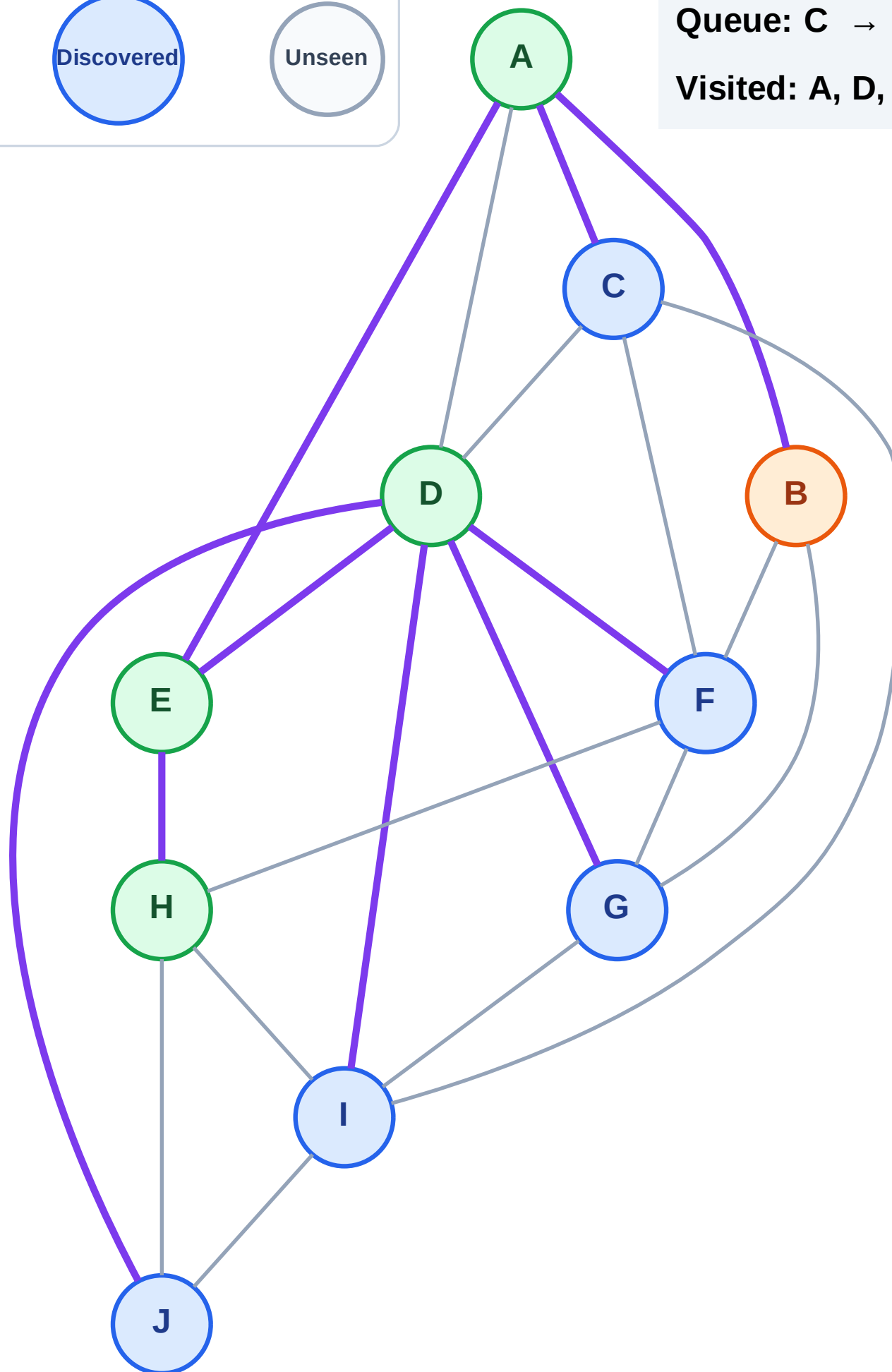
Step 10: Process node B

Legend



Queue: C → F → G → I → J

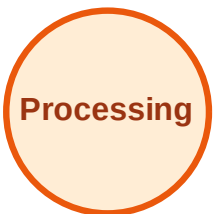
Visited: A, D, E, H



BFS Traversal

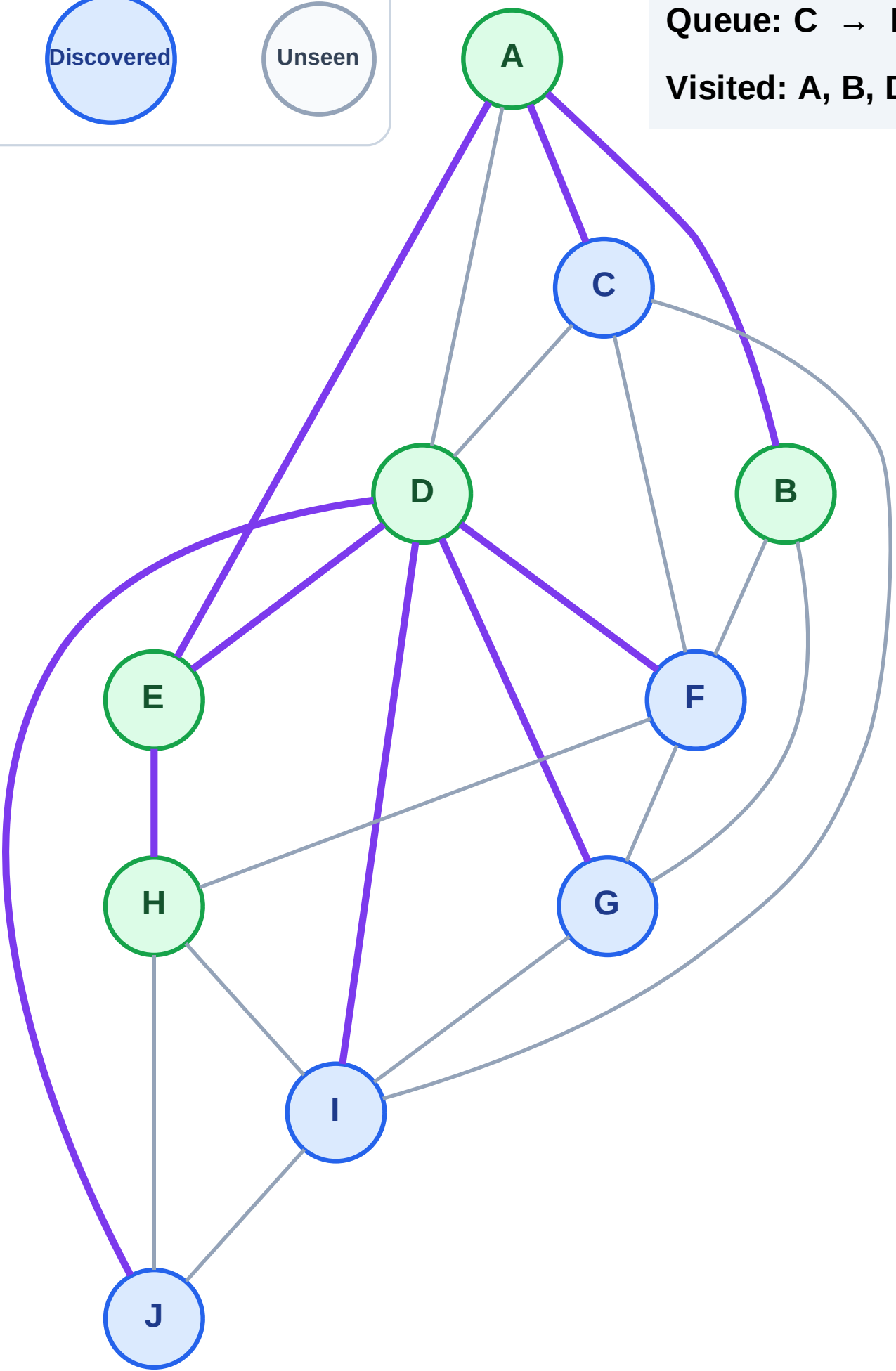
Step 11: No new neighbors from B

Legend



Queue: C → F → G → I → J

Visited: A, B, D, E, H



BFS Traversal

Step 12: Process node C

Legend

Complete

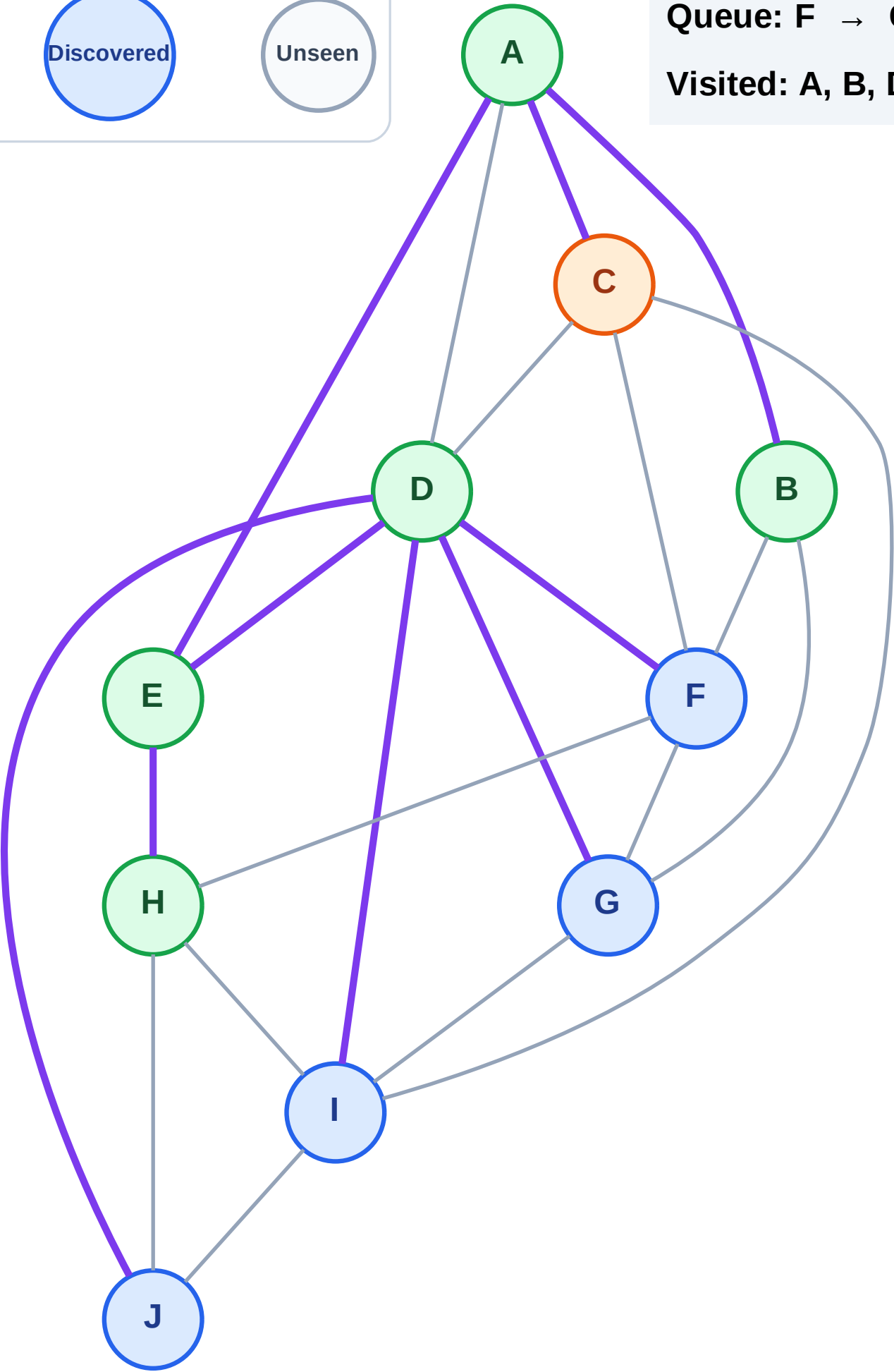
Processing

Discovered

Unseen

Queue: F → G → I → J

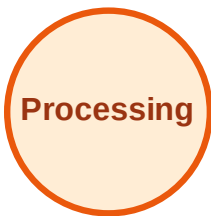
Visited: A, B, D, E, H



BFS Traversal

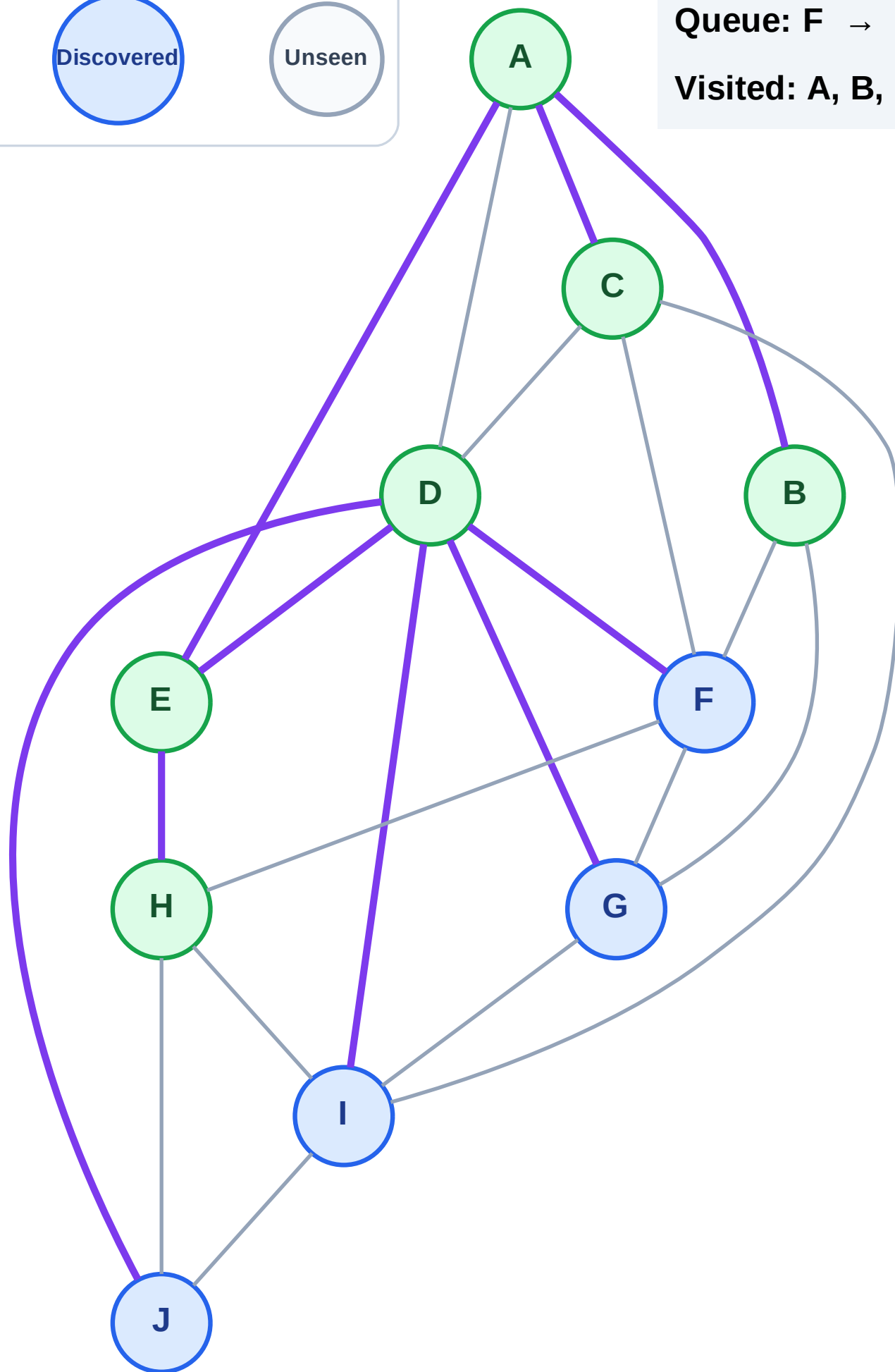
Step 13: No new neighbors from C

Legend



Queue: F → G → I → J

Visited: A, B, C, D, E, H



BFS Traversal

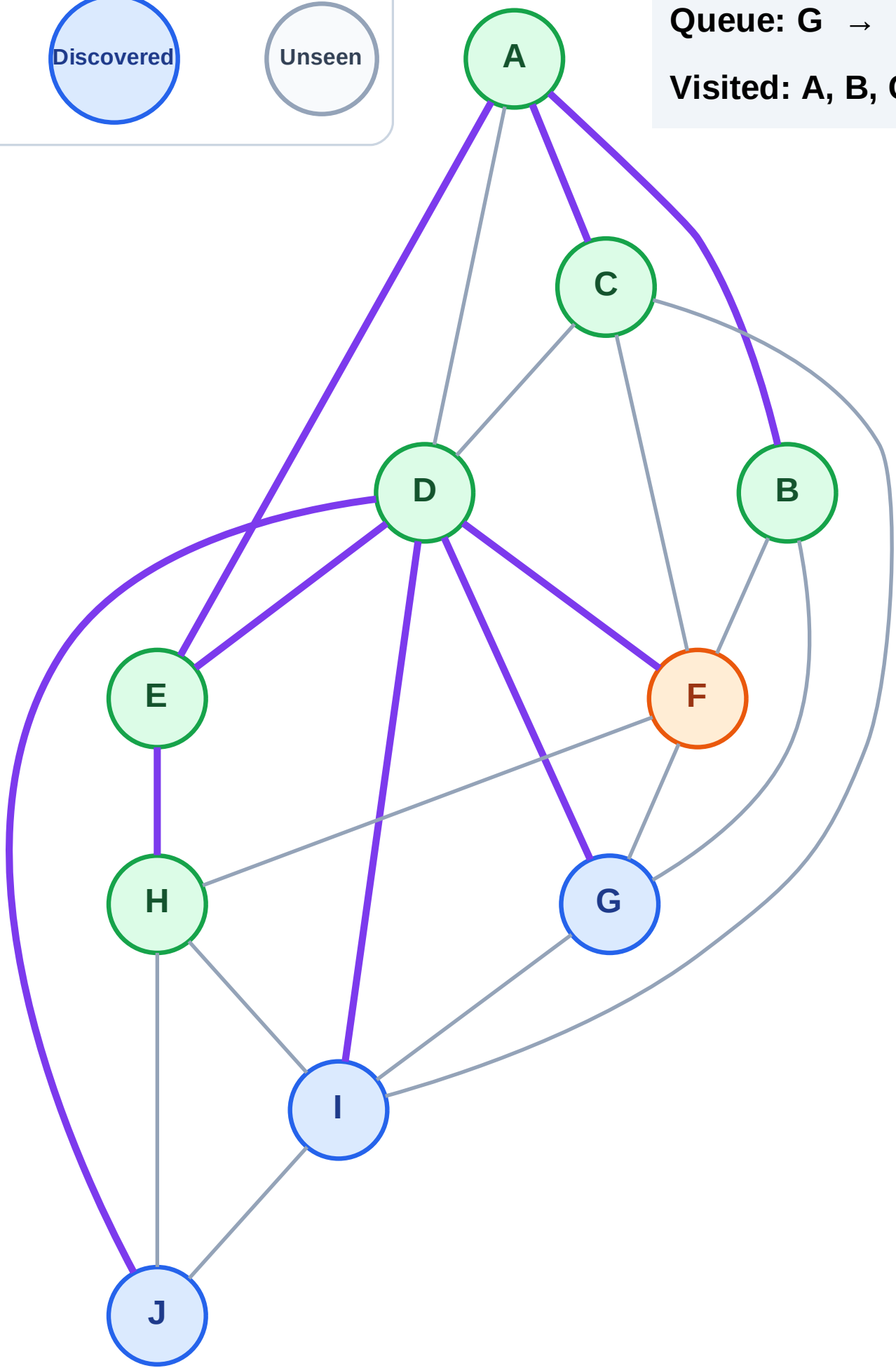
Step 14: Process node F

Legend



Queue: G → I → J

Visited: A, B, C, D, E, H

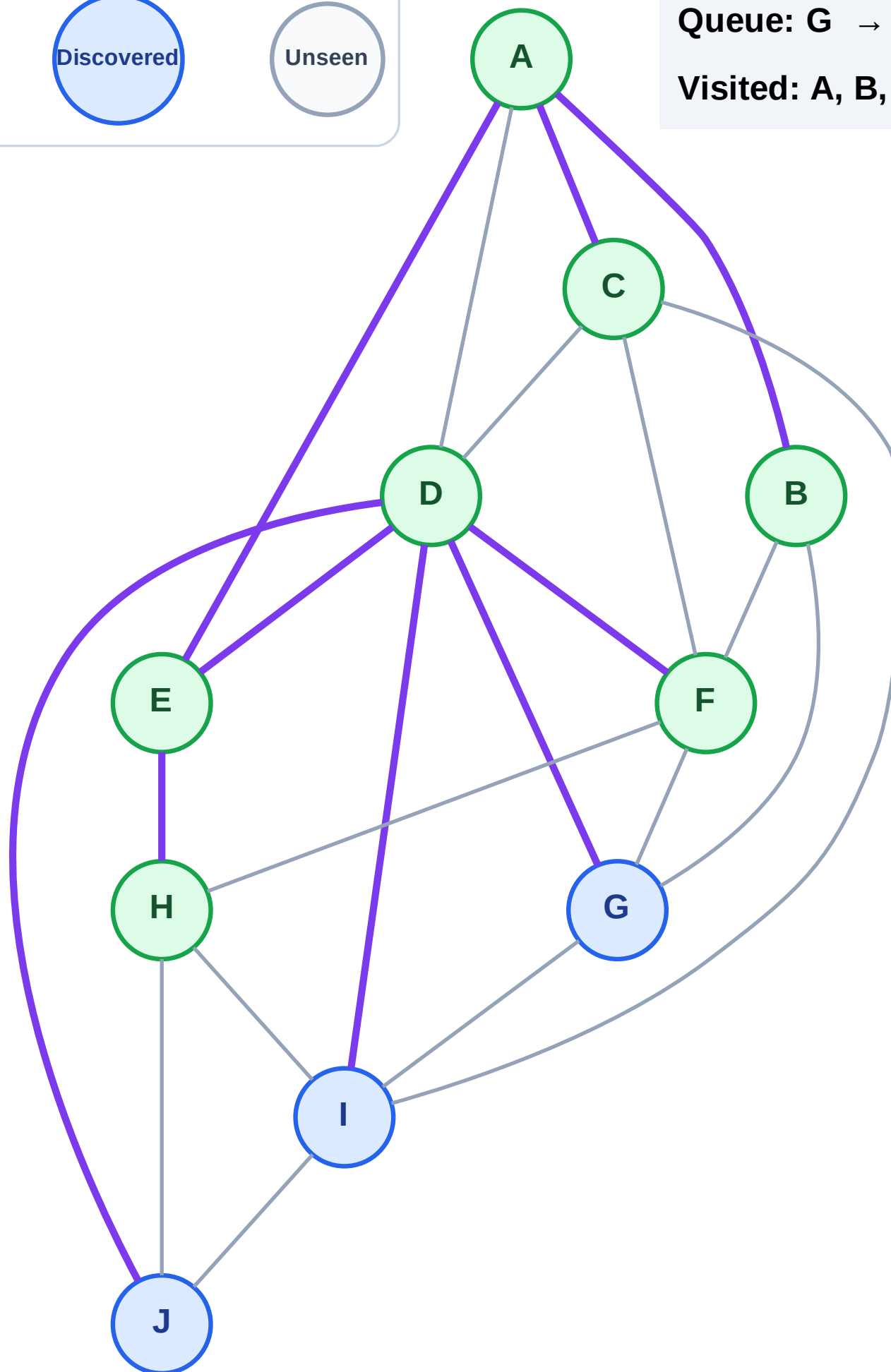


Step 15: No new neighbors from F

Legend



Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node G

Legend

Complete

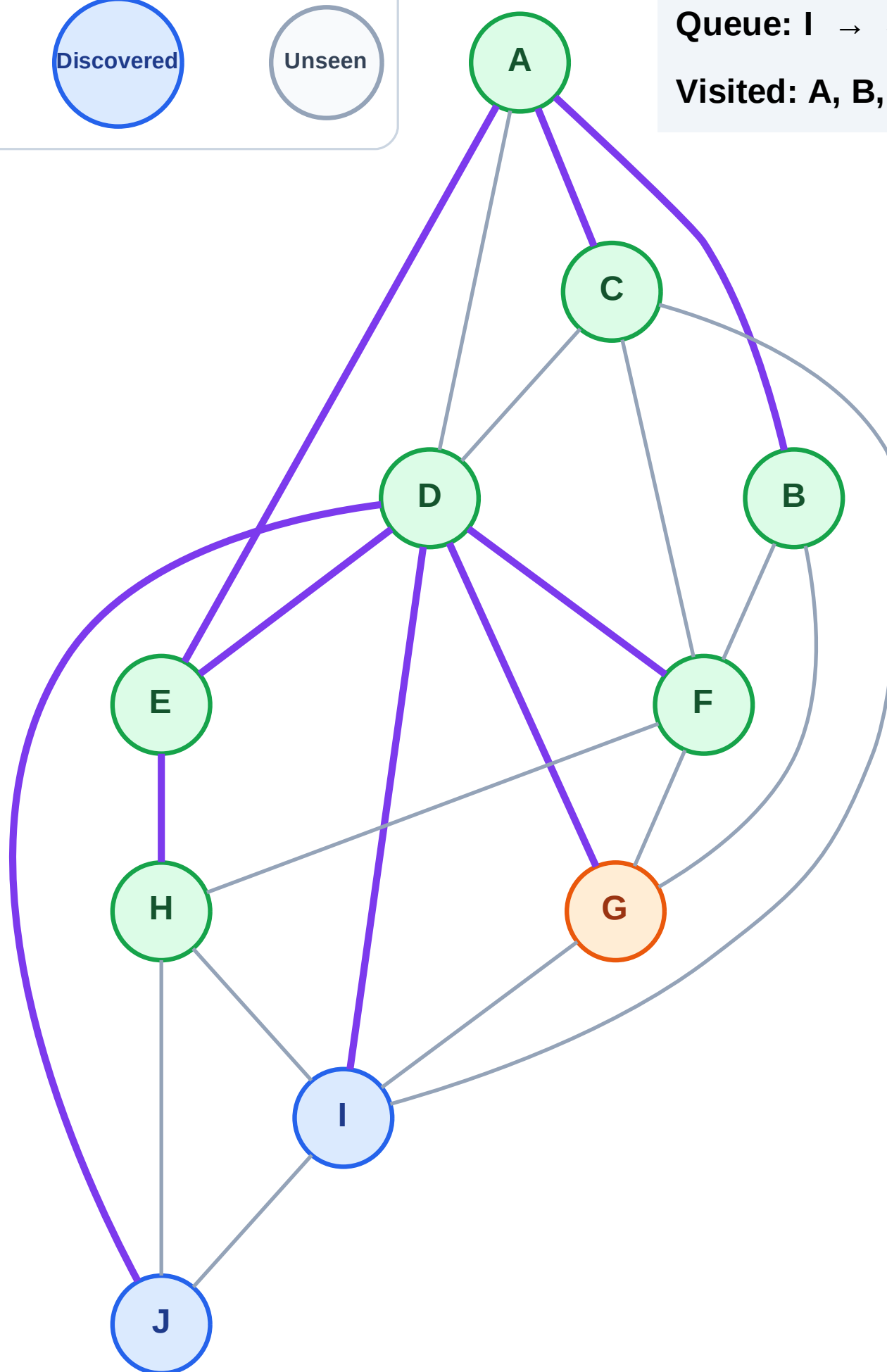
Processing

Discovered

Unseen

Queue: I → J

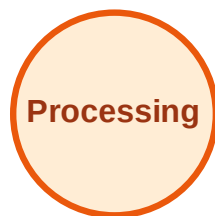
Visited: A, B, C, D, E, F, H



BFS Traversal

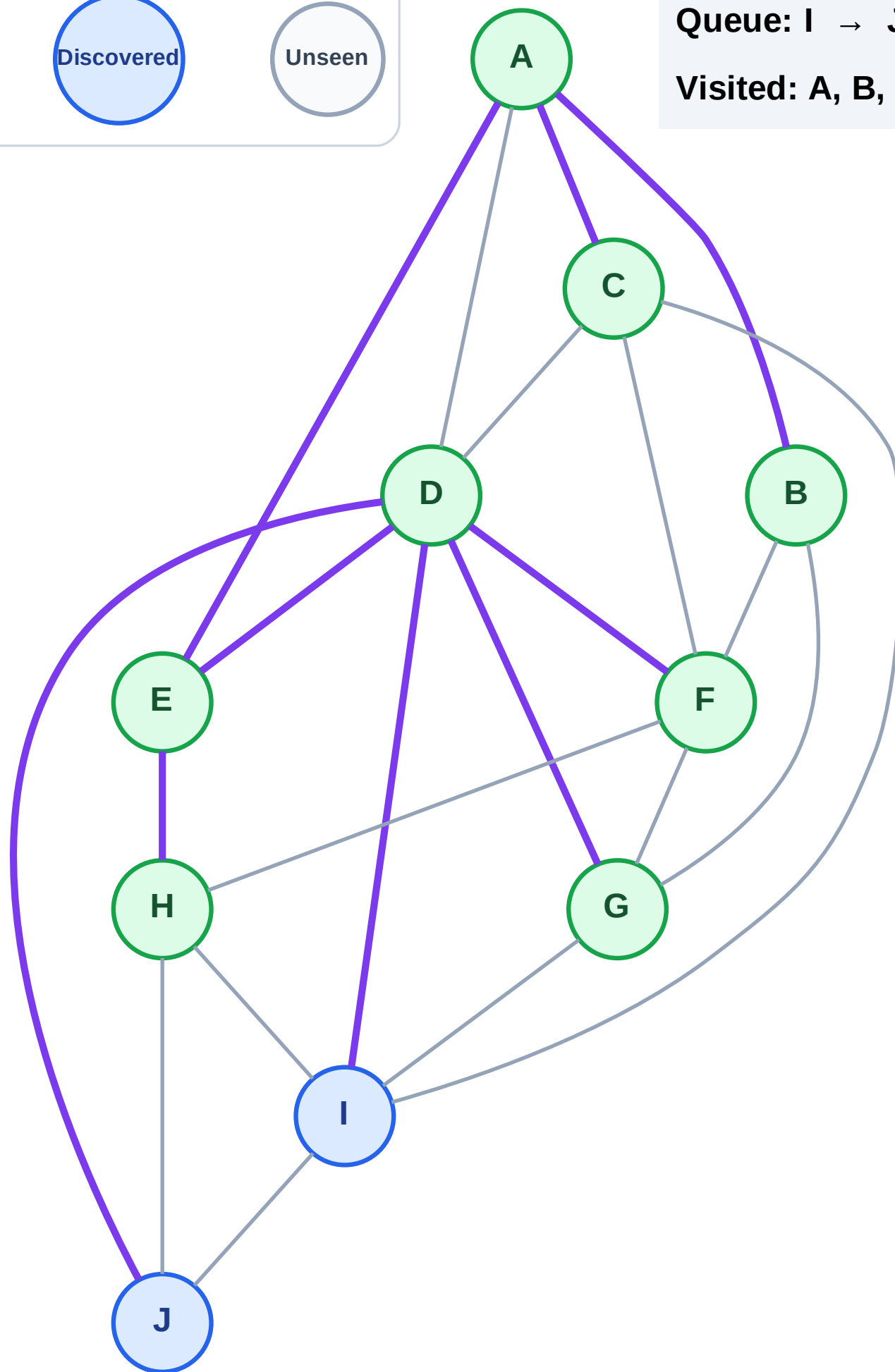
Step 17: No new neighbors from G

Legend



Queue: I → J

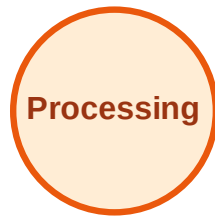
Visited: A, B, C, D, E, F, G, H



BFS Traversal

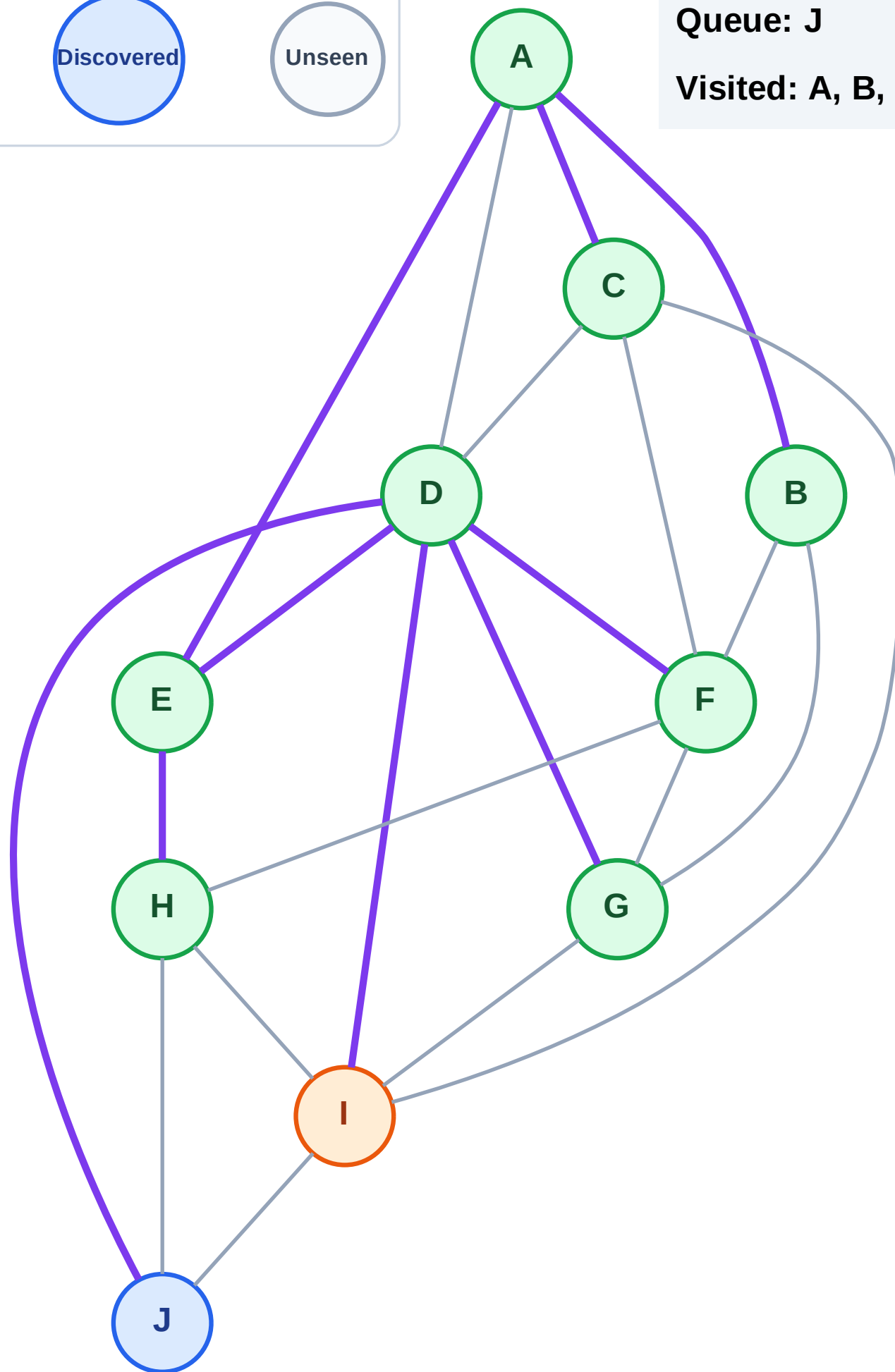
Step 18: Process node I

Legend



Queue: J

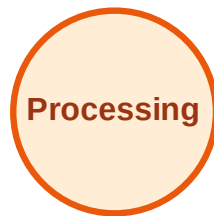
Visited: A, B, C, D, E, F, G, H



BFS Traversal

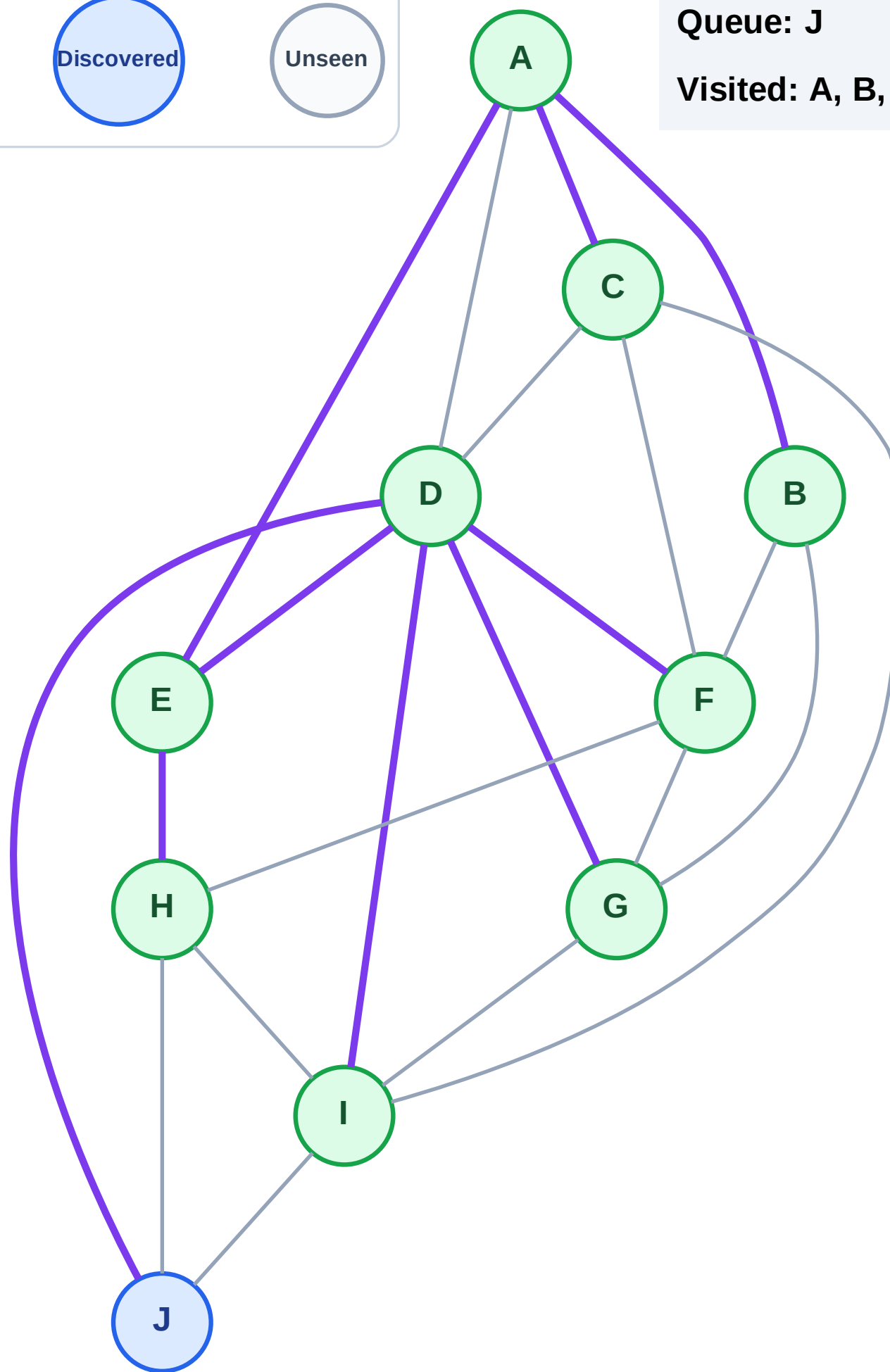
Step 19: No new neighbors from I

Legend



Queue: J

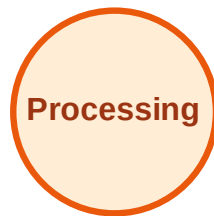
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

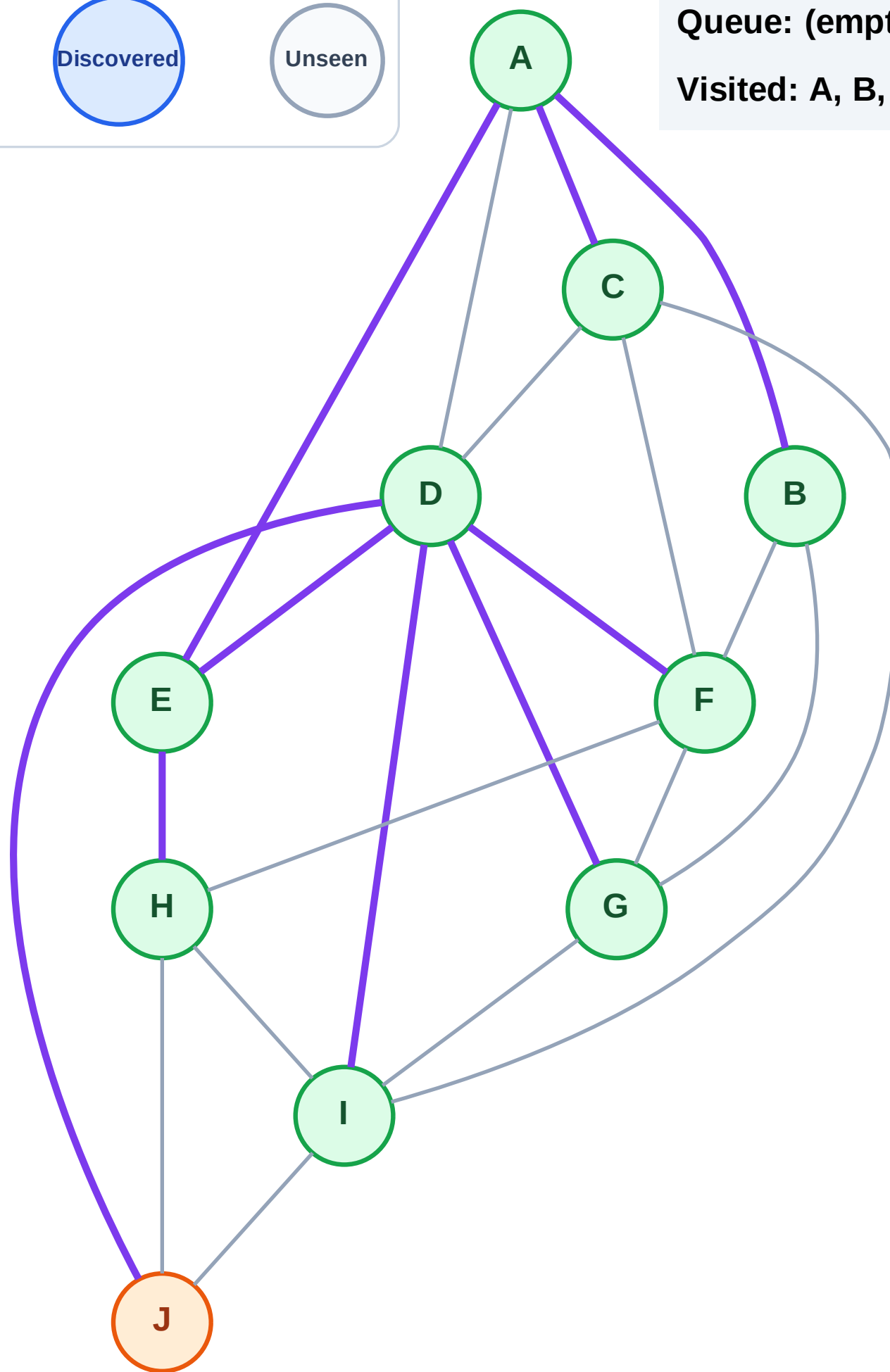
Step 20: Process node J

Legend



Queue: (empty)

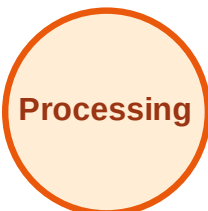
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

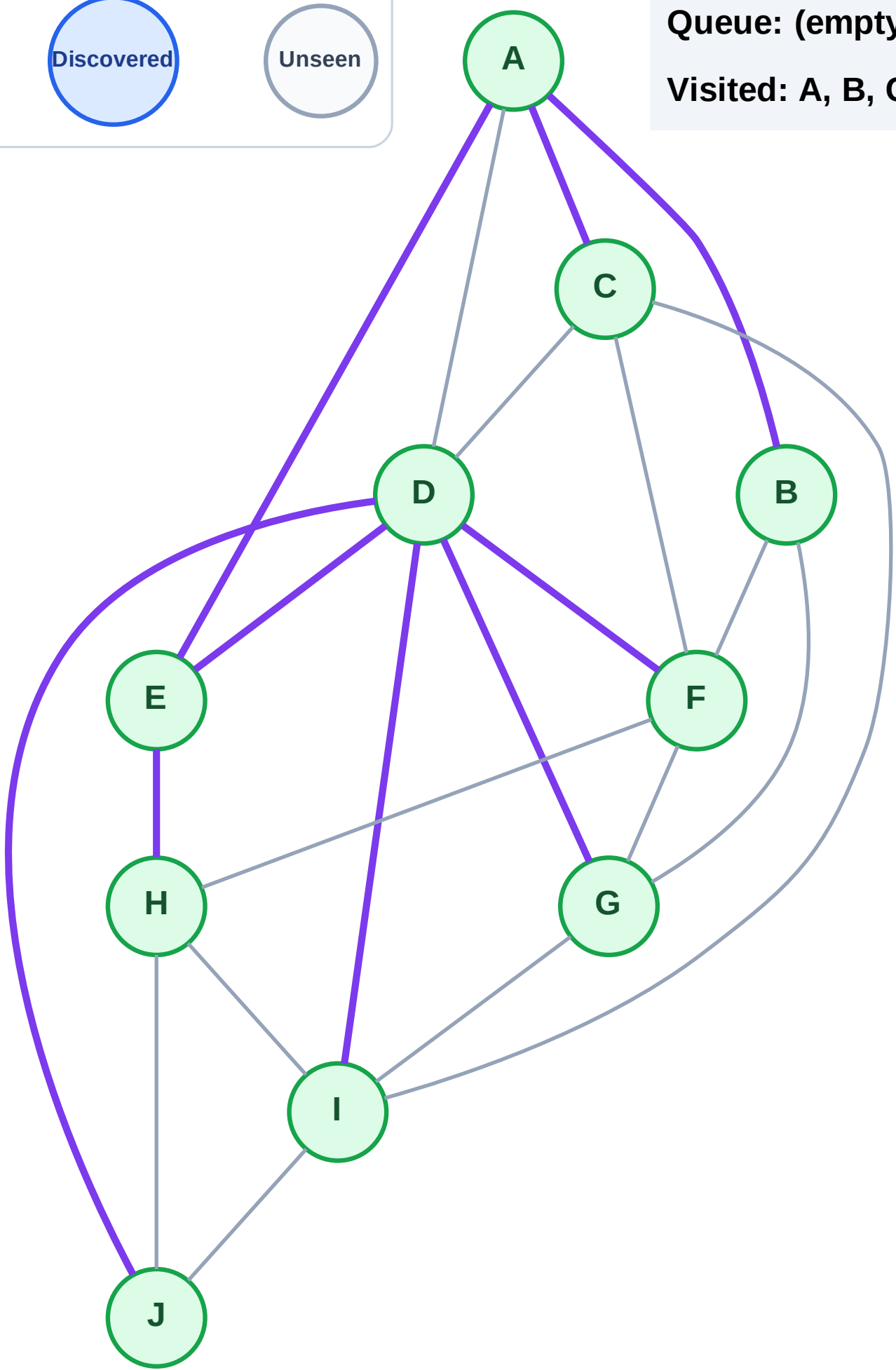
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

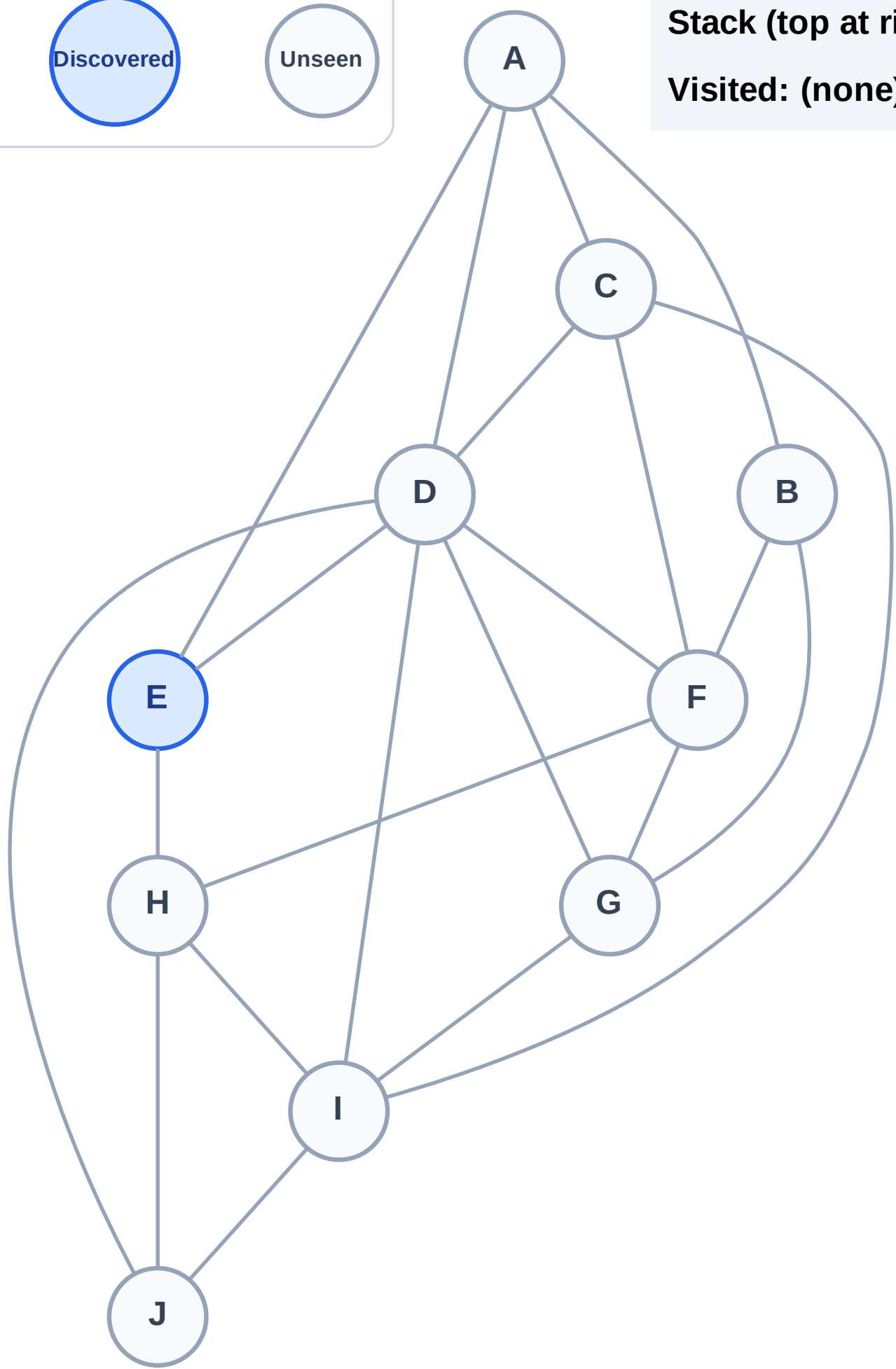
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

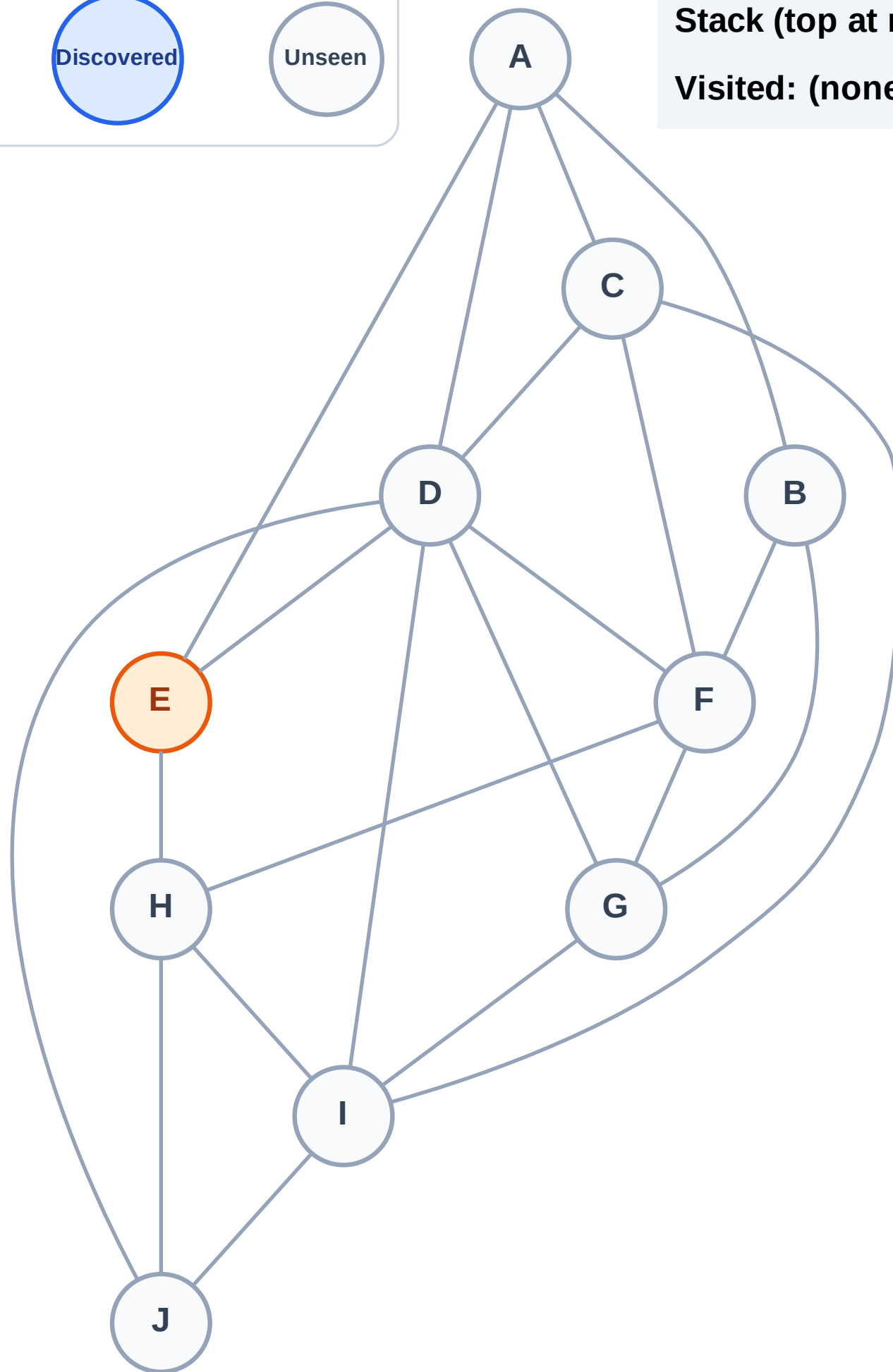
Processing

Discovered

Unseen

Stack (top at right): (empty)

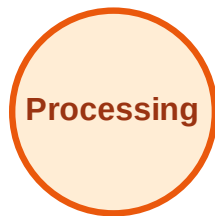
Visited: (none)



DFS Traversal

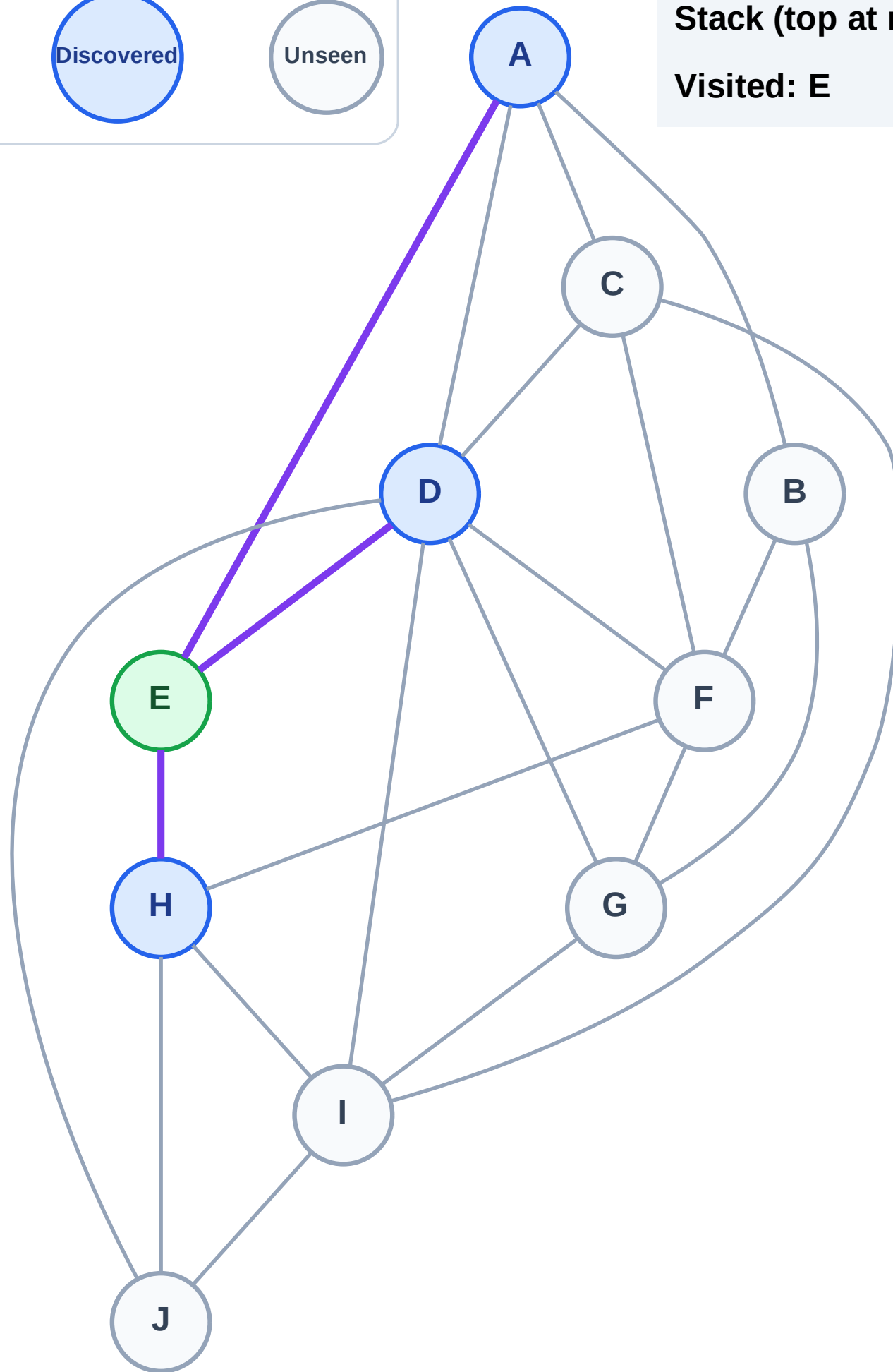
Step 3: Discover H, D, A from E

Legend



Stack (top at right): H | D | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

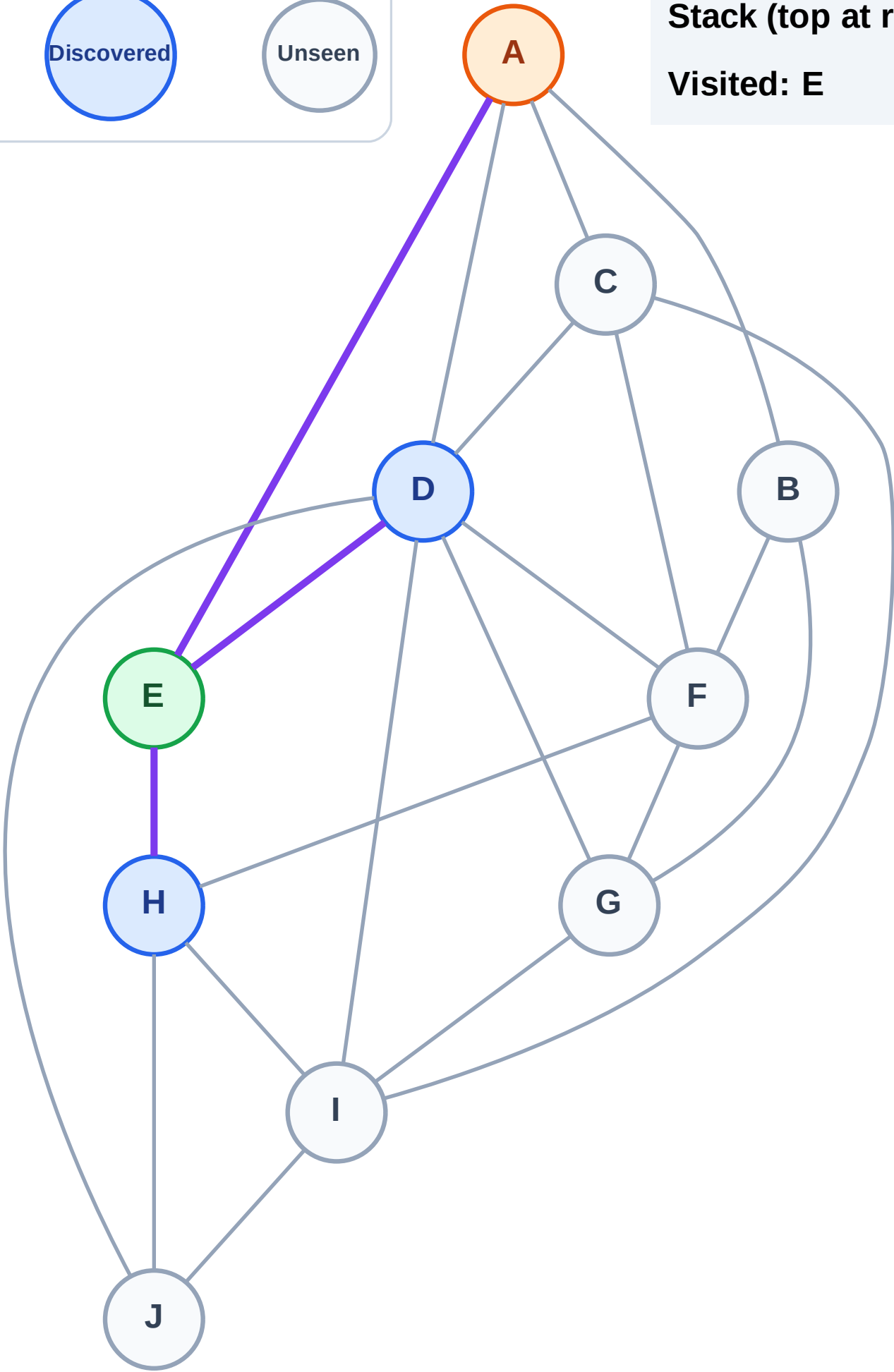
Processing

Discovered

Unseen

Stack (top at right): H | D

Visited: E



DFS Traversal

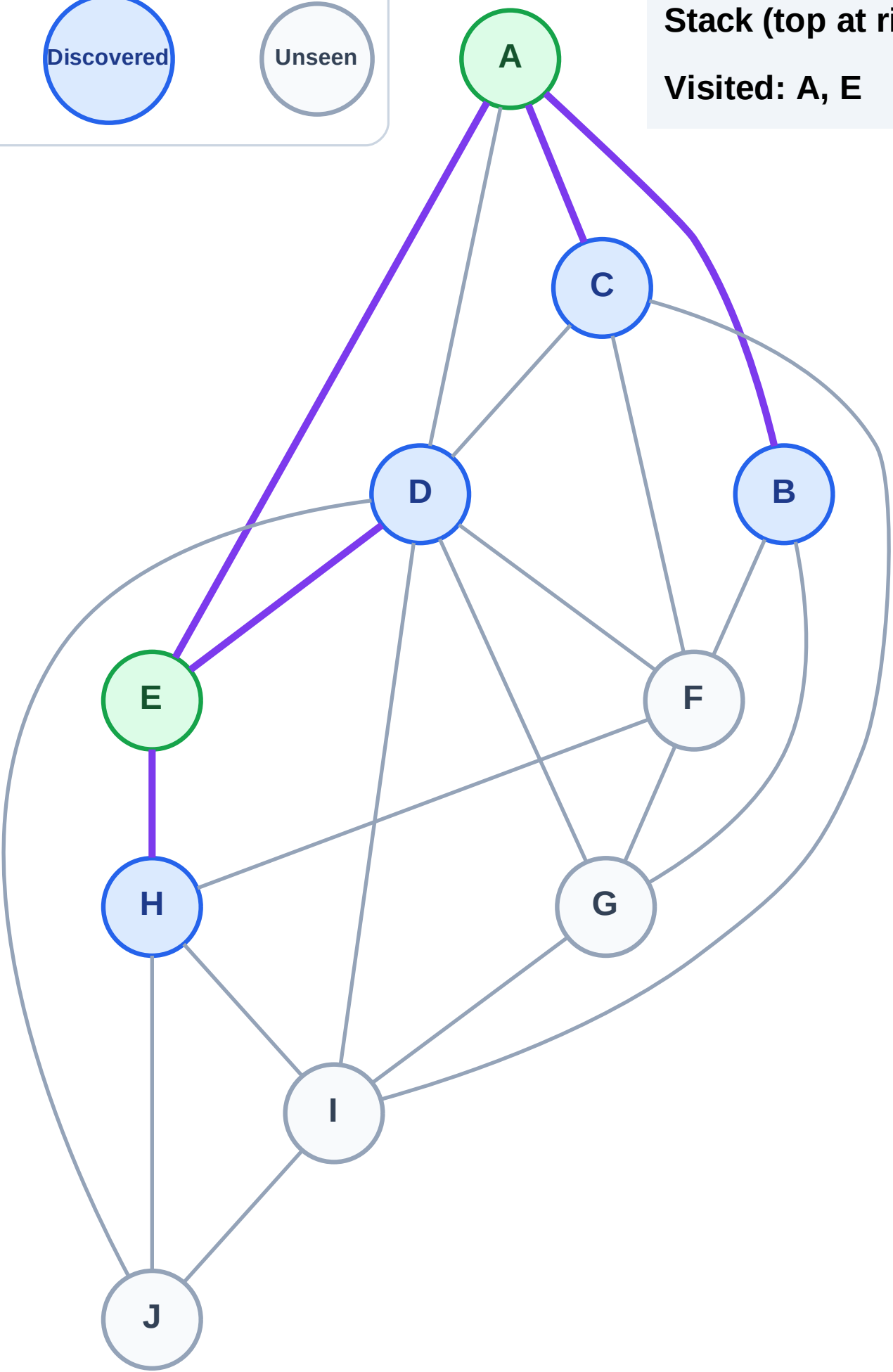
Step 5: Discover C, B from A

Legend



Stack (top at right): H | D | C | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

Complete

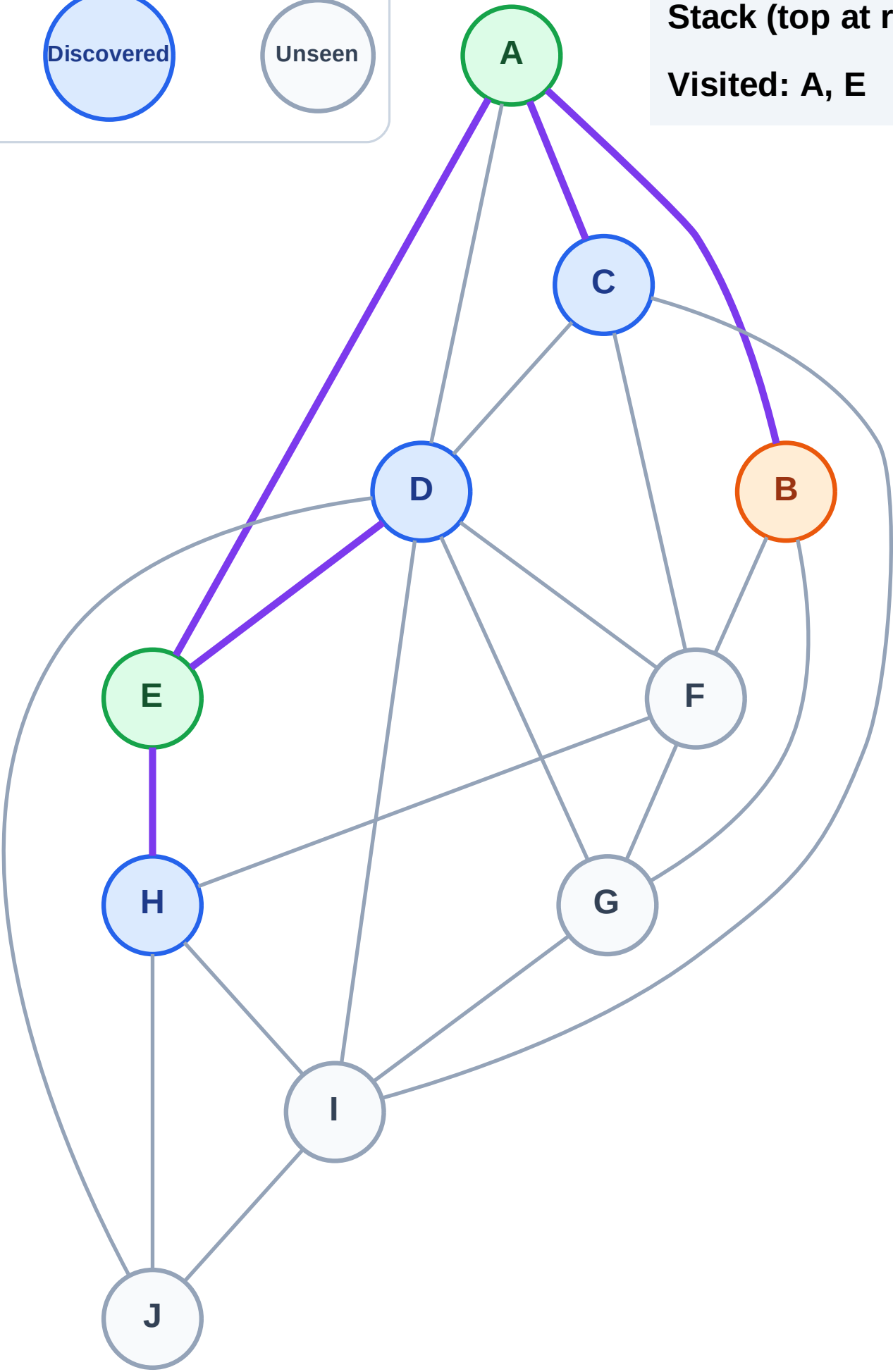
Processing

Discovered

Unseen

Stack (top at right): H | D | C

Visited: A, E



DFS Traversal

Step 7: Discover G, F from B

Legend

Complete

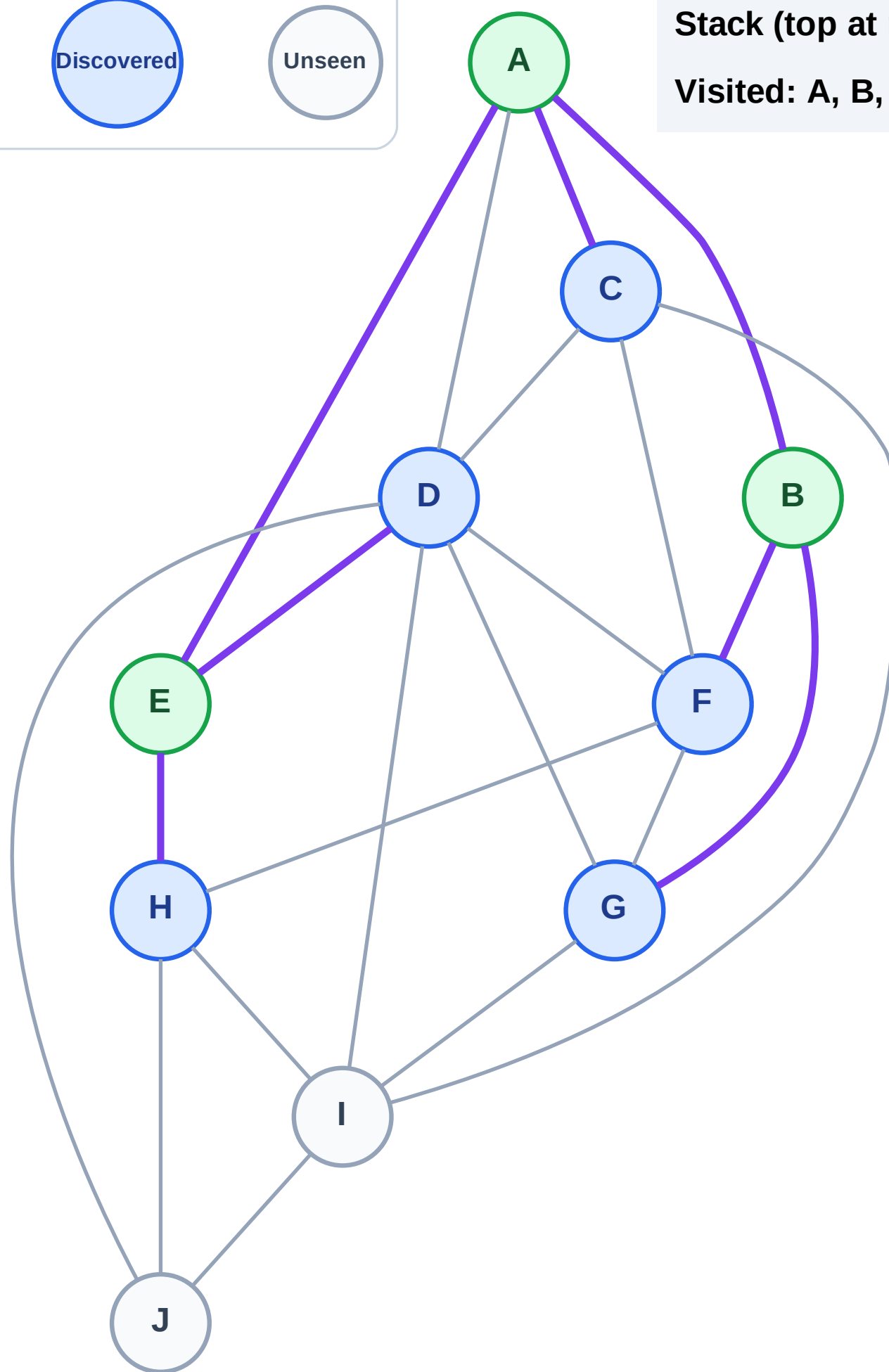
Processing

Discovered

Unseen

Stack (top at right): H | D | C | G | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

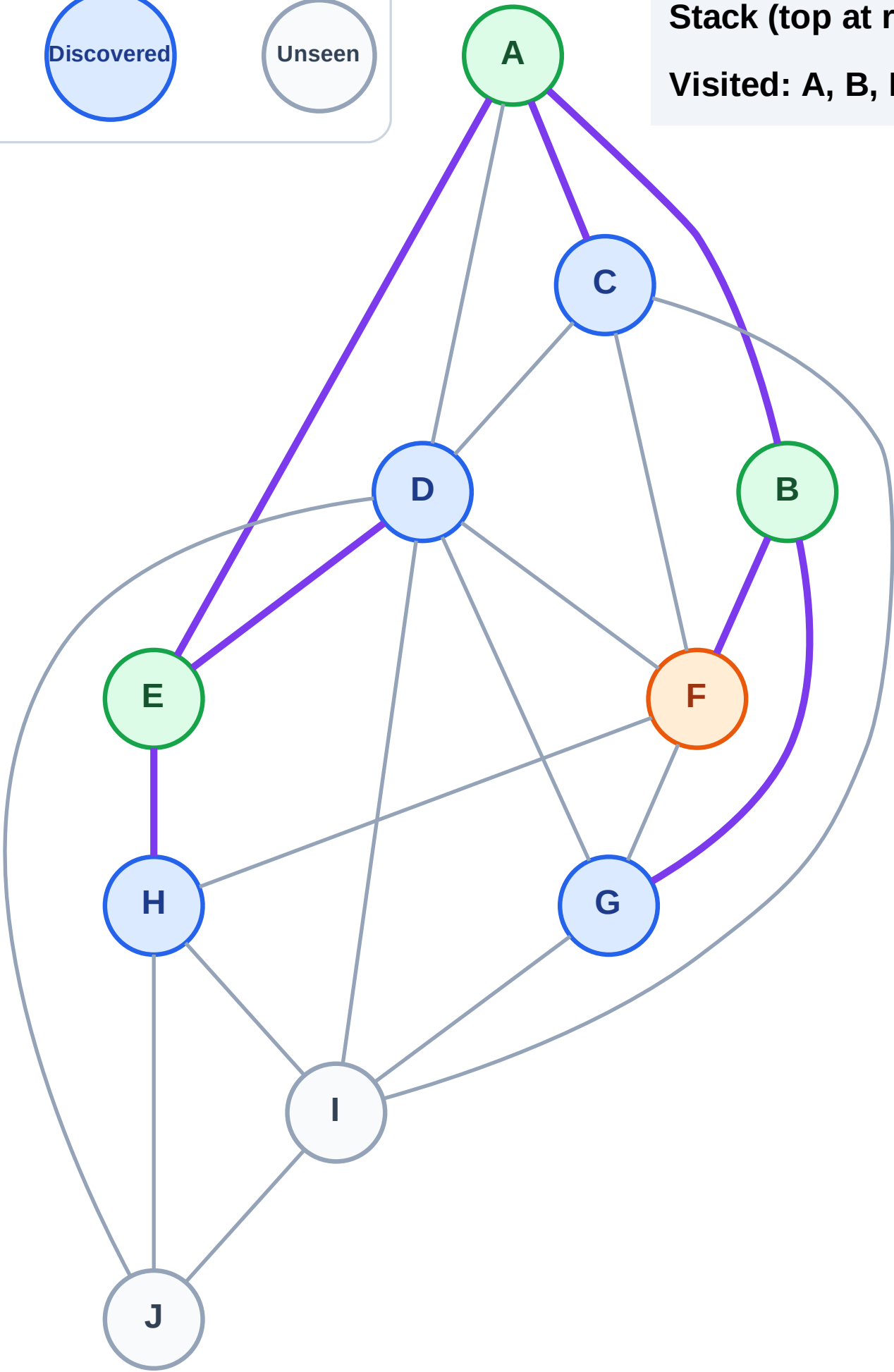
Processing

Discovered

Unseen

Stack (top at right): H | D | C | G

Visited: A, B, E



DFS Traversal

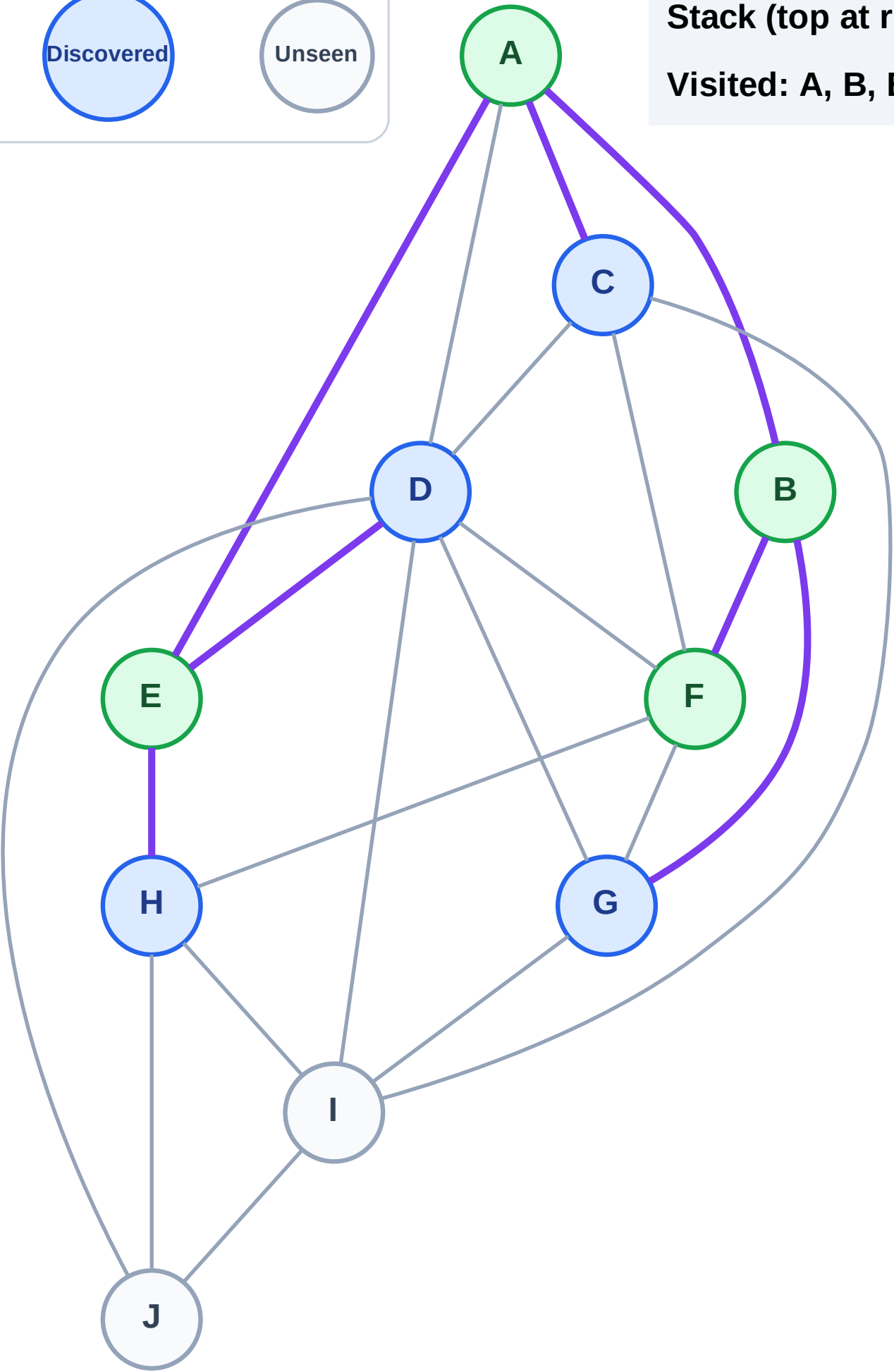
Step 9: No new neighbors from F

Legend



Stack (top at right): H | D | C | G

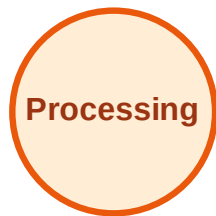
Visited: A, B, E, F



DFS Traversal

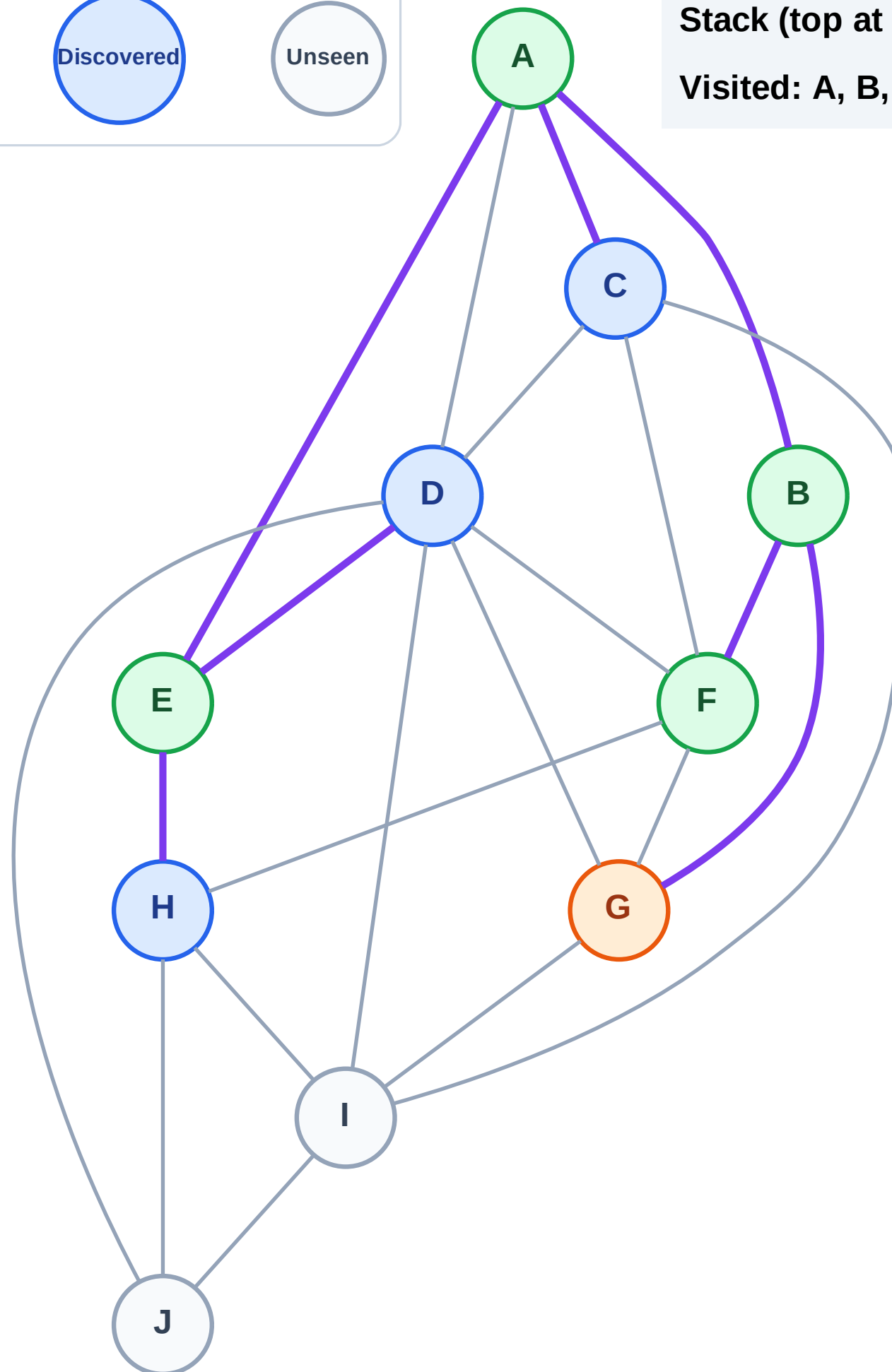
Step 10: Process node G

Legend



Stack (top at right): H | D | C

Visited: A, B, E, F



DFS Traversal

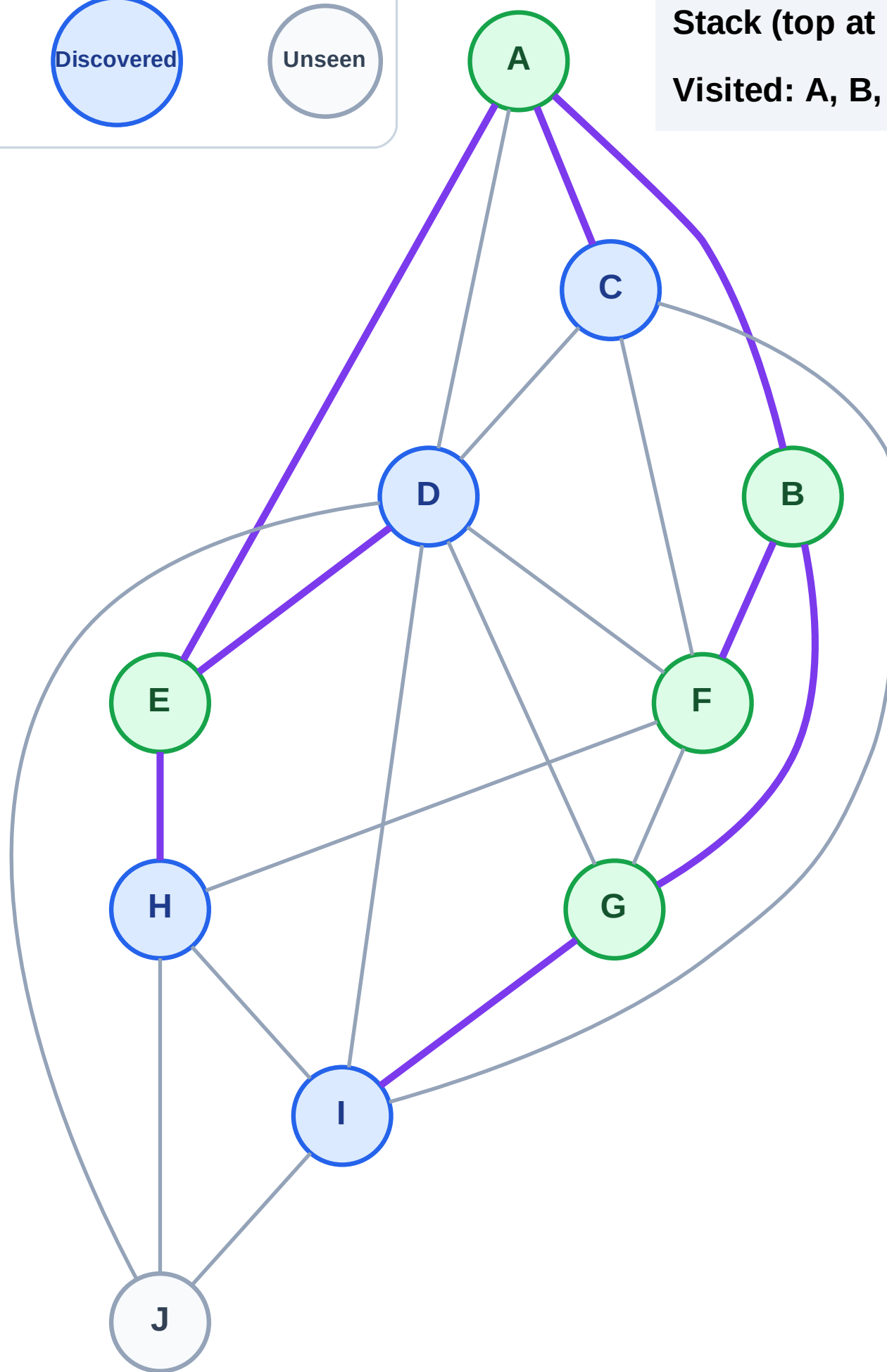
Step 11: Discover I from G

Legend



Stack (top at right): H | D | C | I

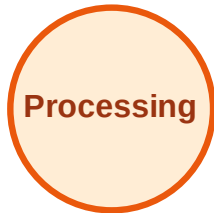
Visited: A, B, E, F, G



DFS Traversal

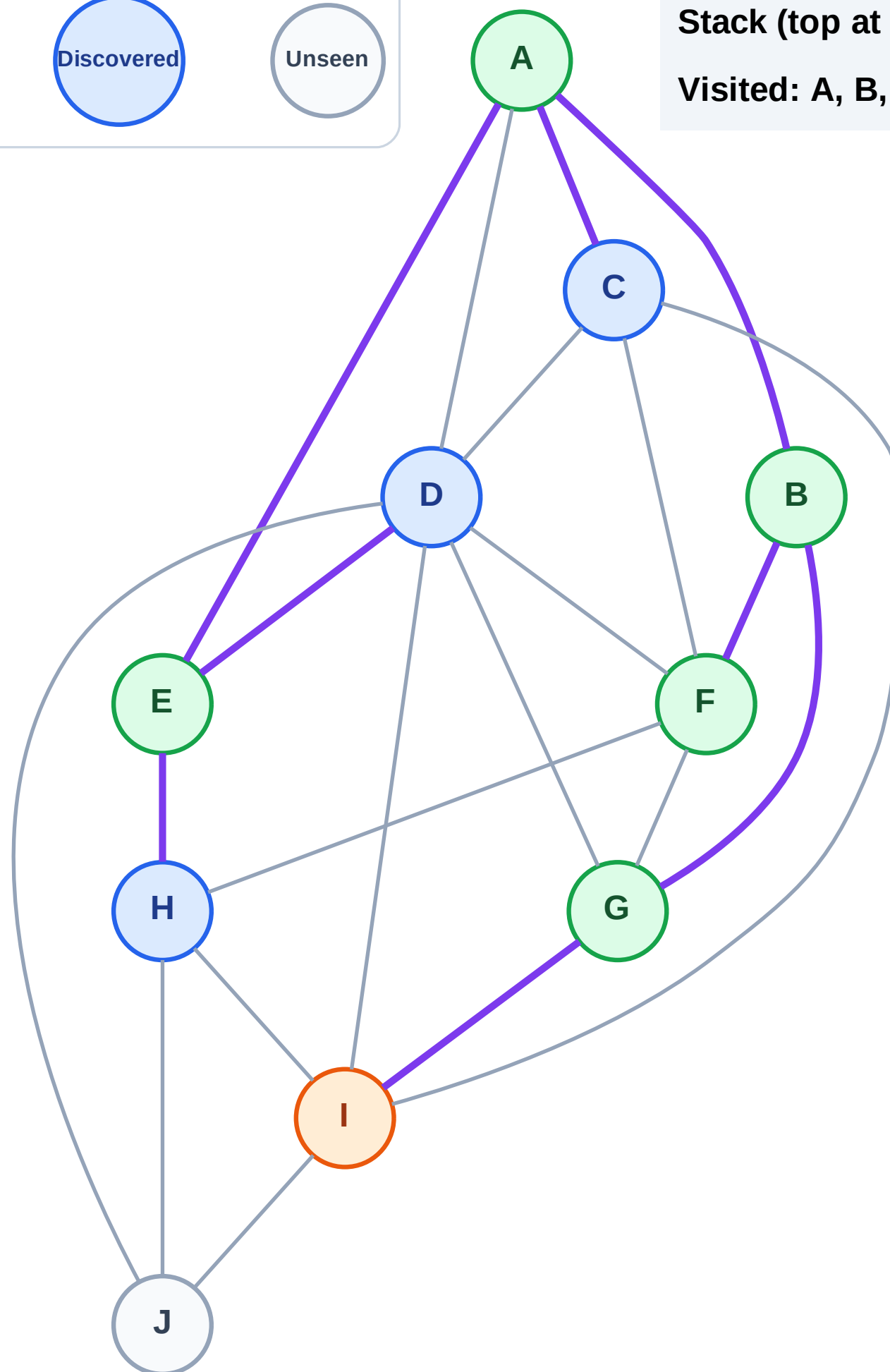
Step 12: Process node I

Legend



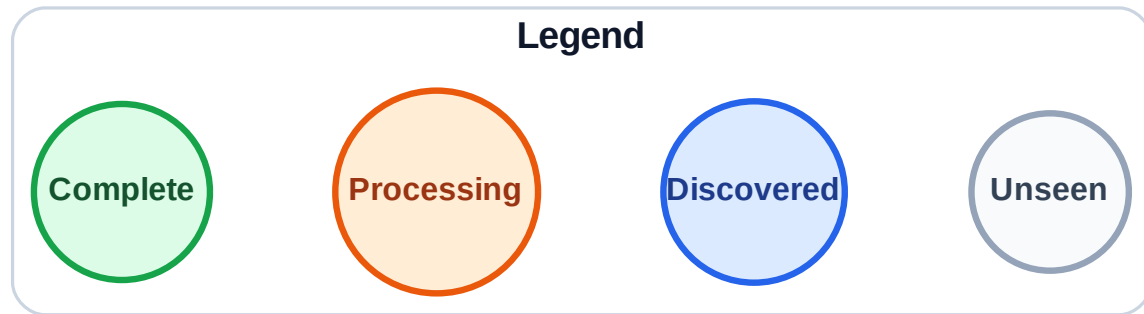
Stack (top at right): H | D | C

Visited: A, B, E, F, G

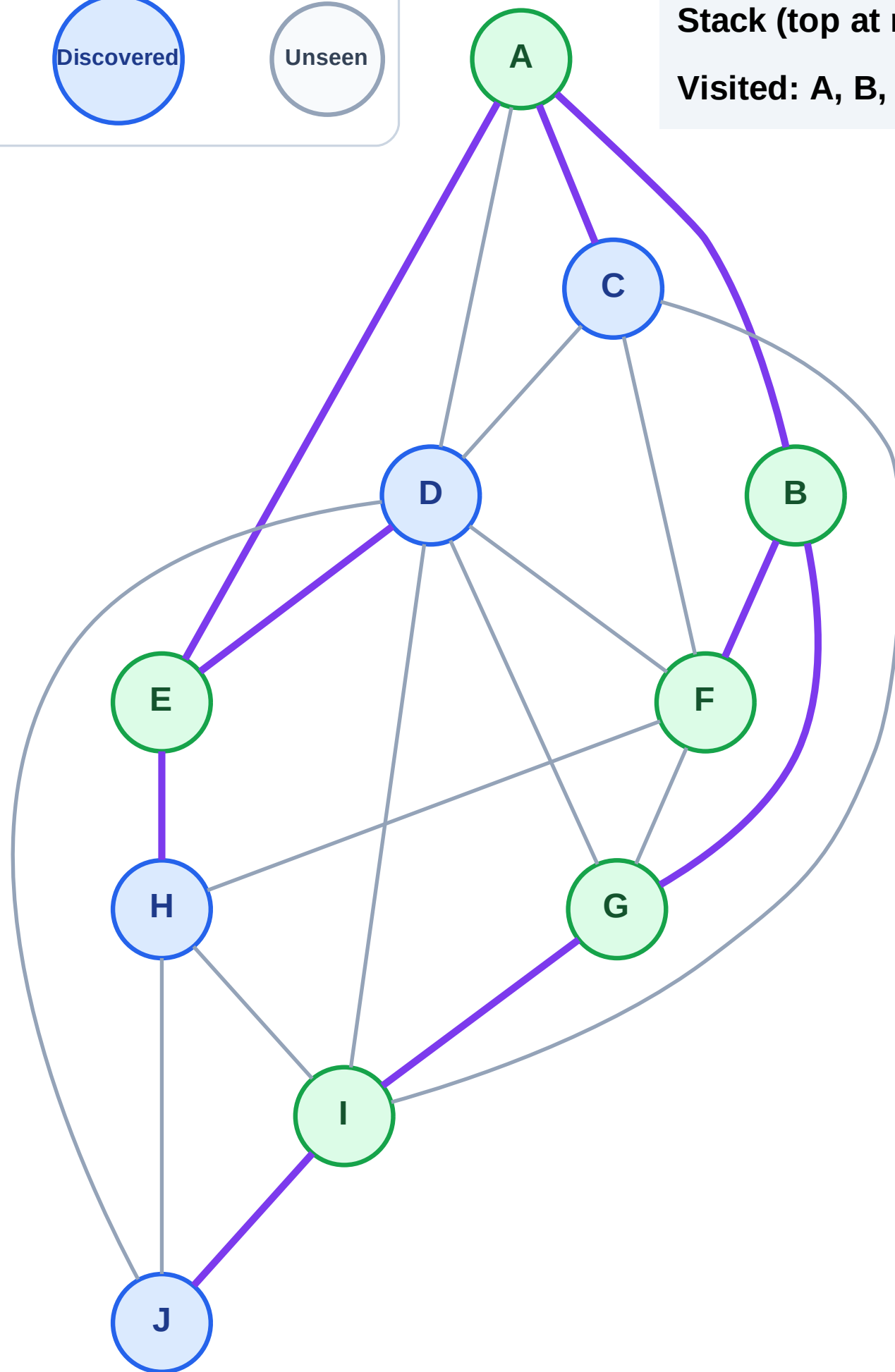


DFS Traversal

Step 13: Discover J from I



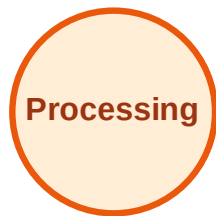
Stack (top at right): H | D | C | J
Visited: A, B, E, F, G, I



DFS Traversal

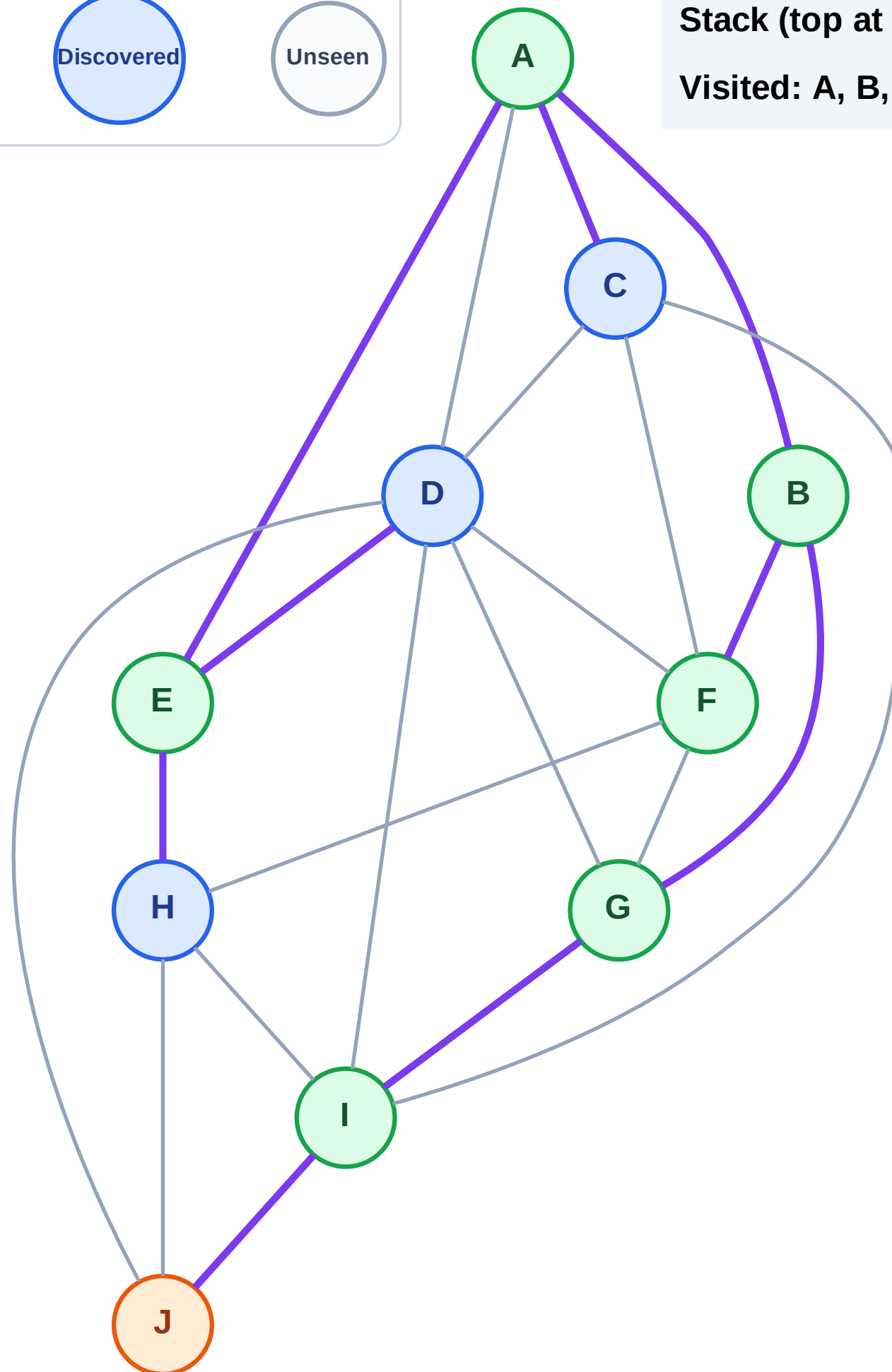
Step 14: Process node J

Legend



Stack (top at right): H | D | C

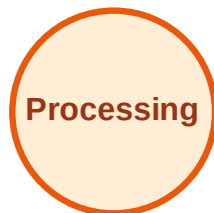
Visited: A, B, E, F, G, I



DFS Traversal

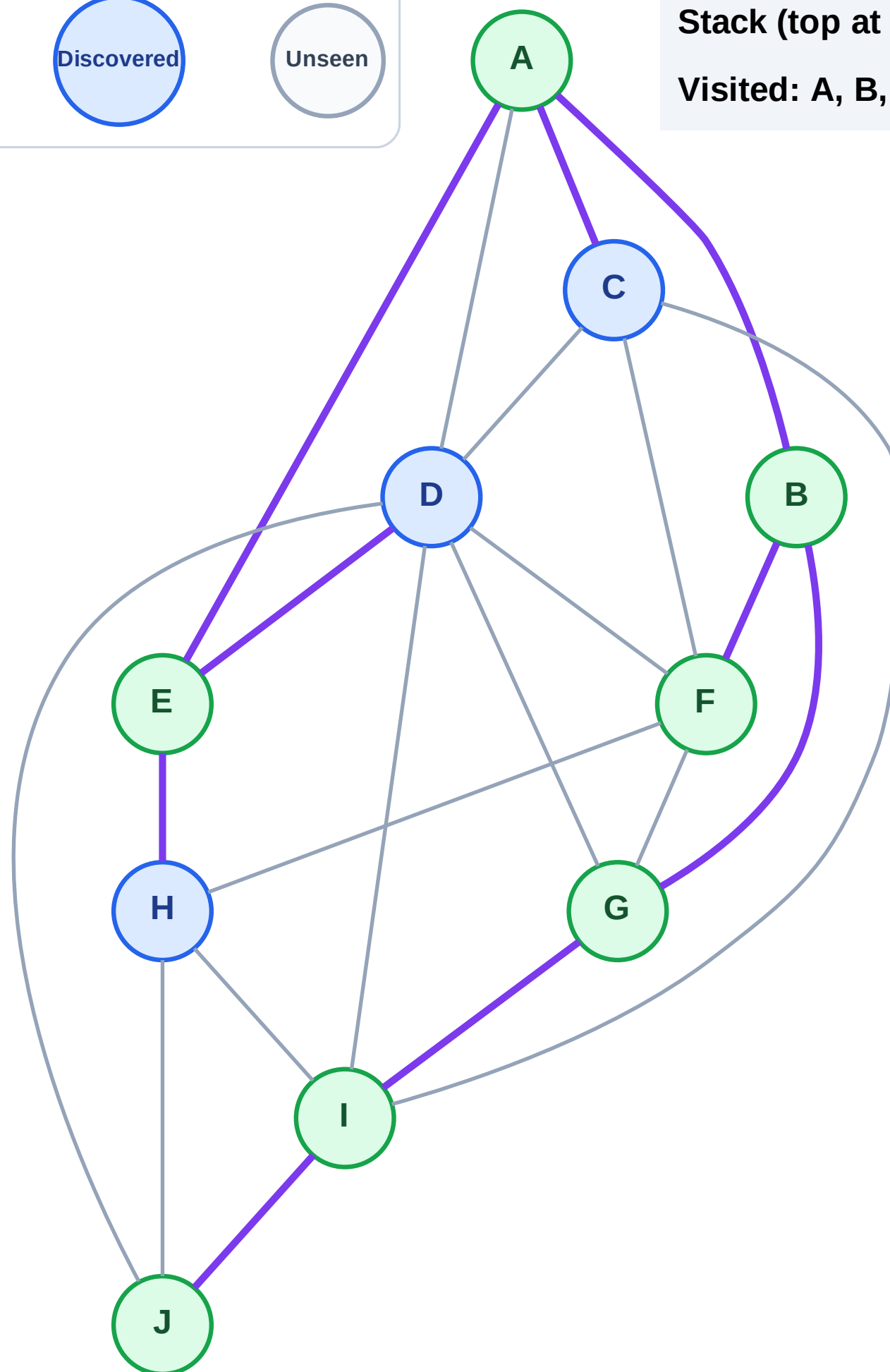
Step 15: No new neighbors from J

Legend



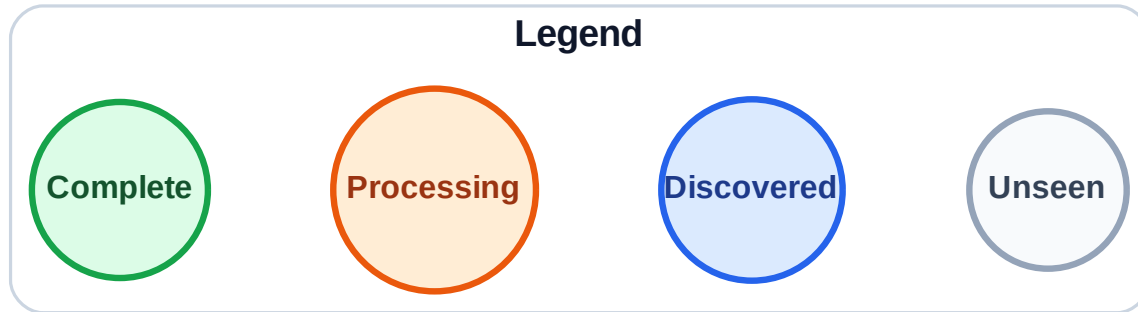
Stack (top at right): H | D | C

Visited: A, B, E, F, G, I, J



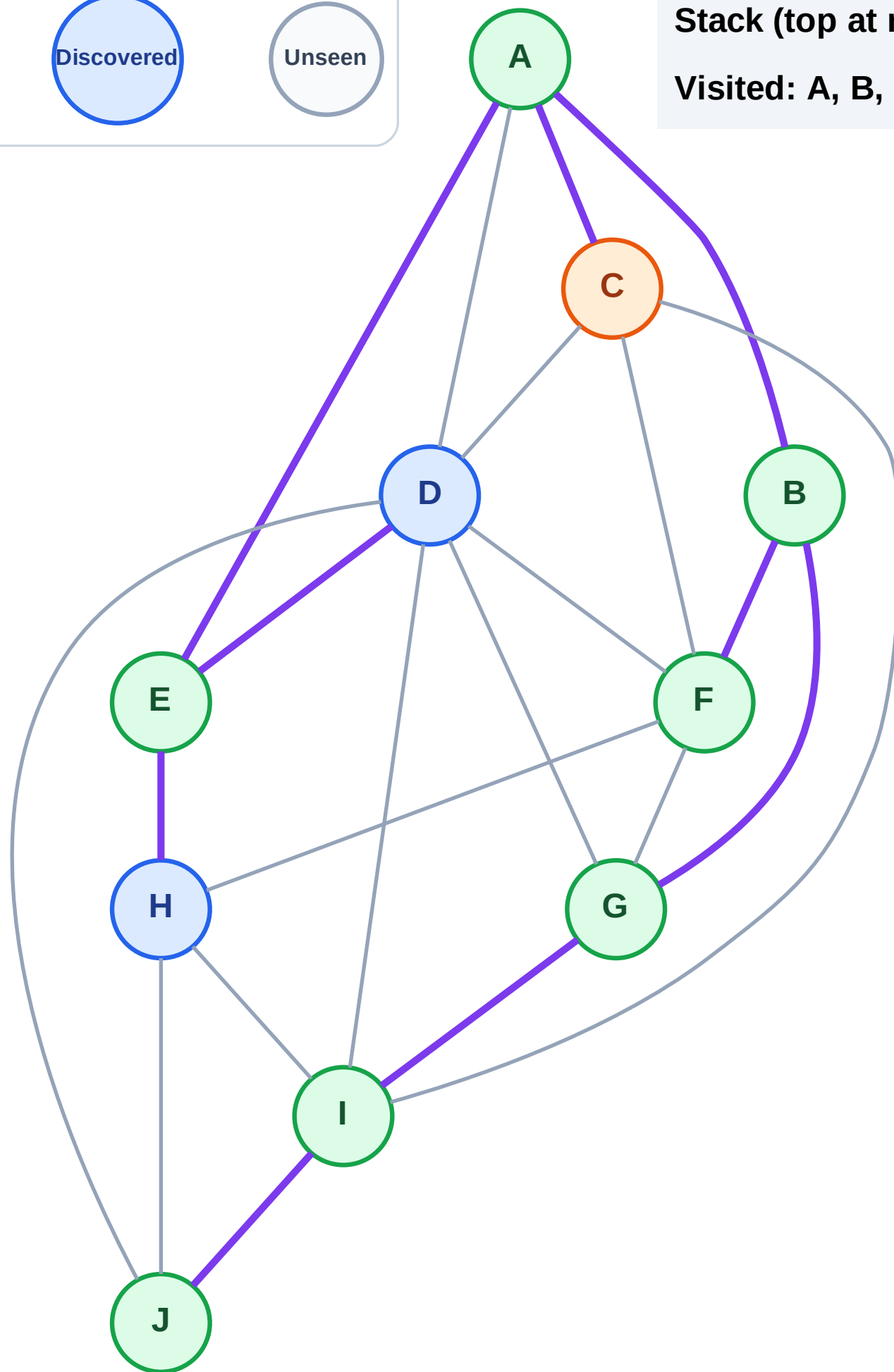
Step 16: Process node C

Step 16: Process node C



Stack (top at right): H | D

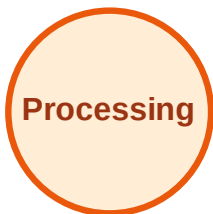
Visited: A, B, E, F, G, I, J



DFS Traversal

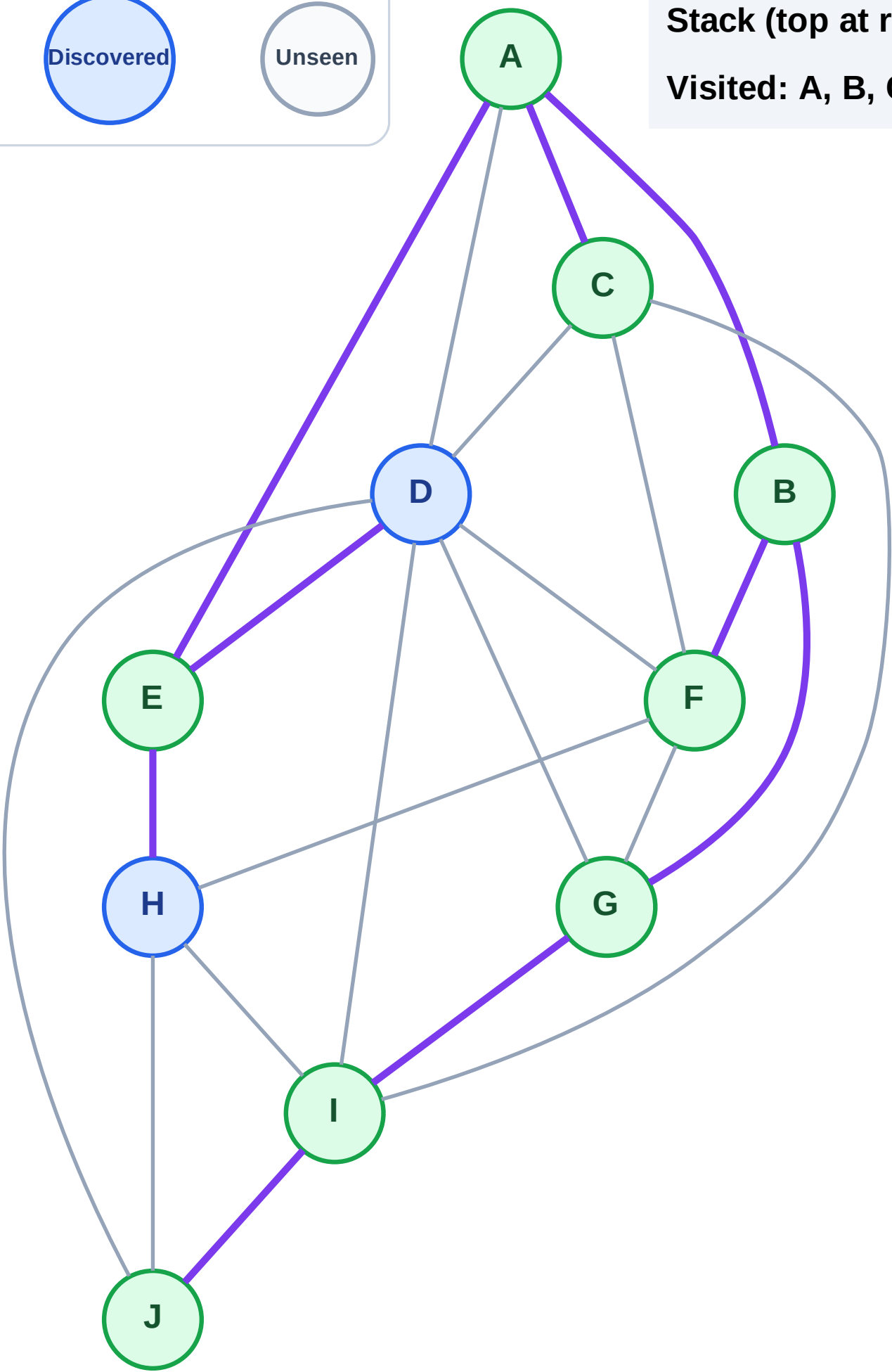
Step 17: No new neighbors from C

Legend



Stack (top at right): H | D

Visited: A, B, C, E, F, G, I, J



DFS Traversal

Step 18: Process node D

Legend

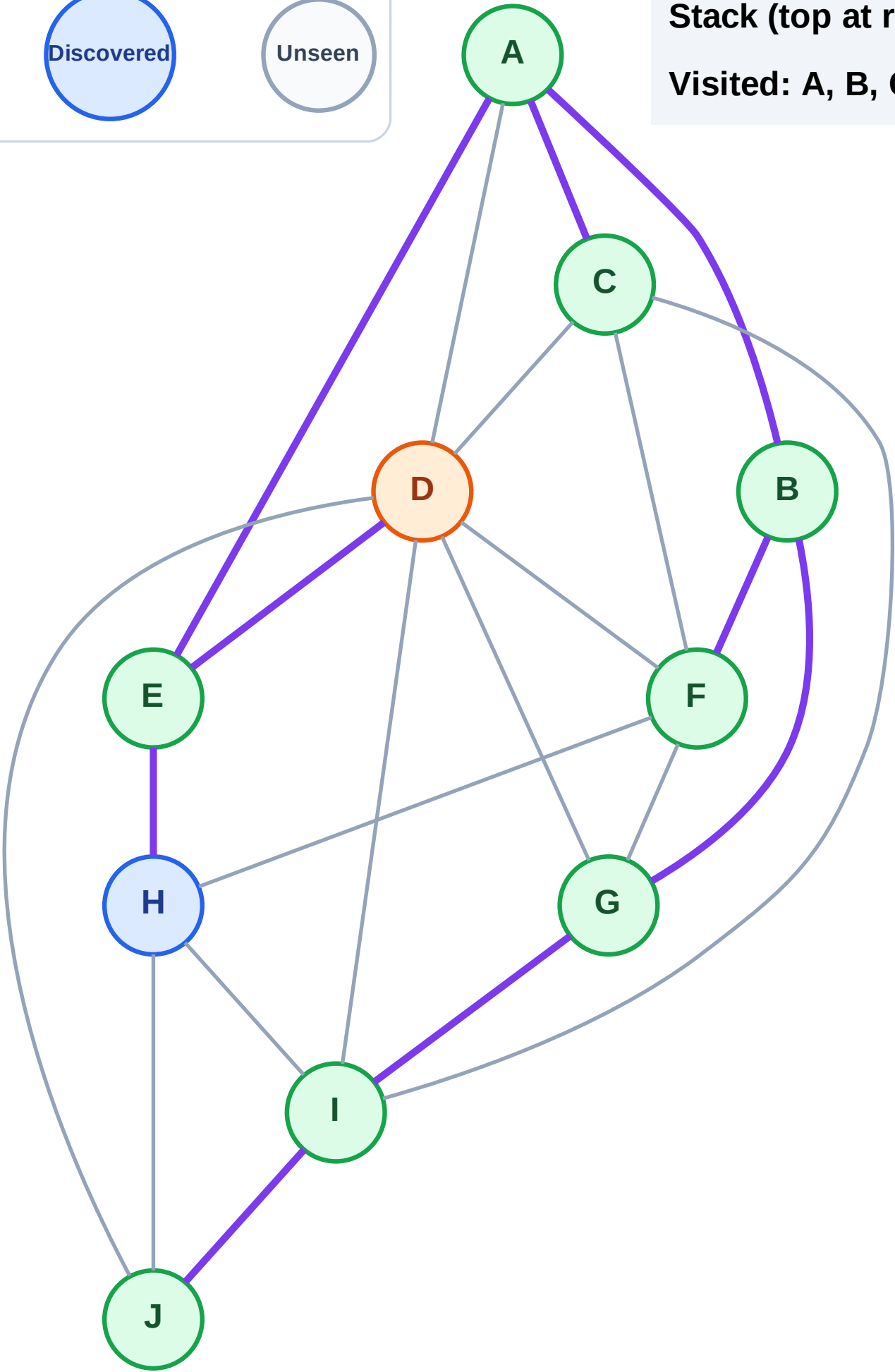
Complete

Processing

Discovered

Unseen

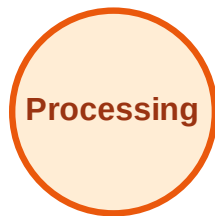
Stack (top at right): H
Visited: A, B, C, E, F, G, I, J



DFS Traversal

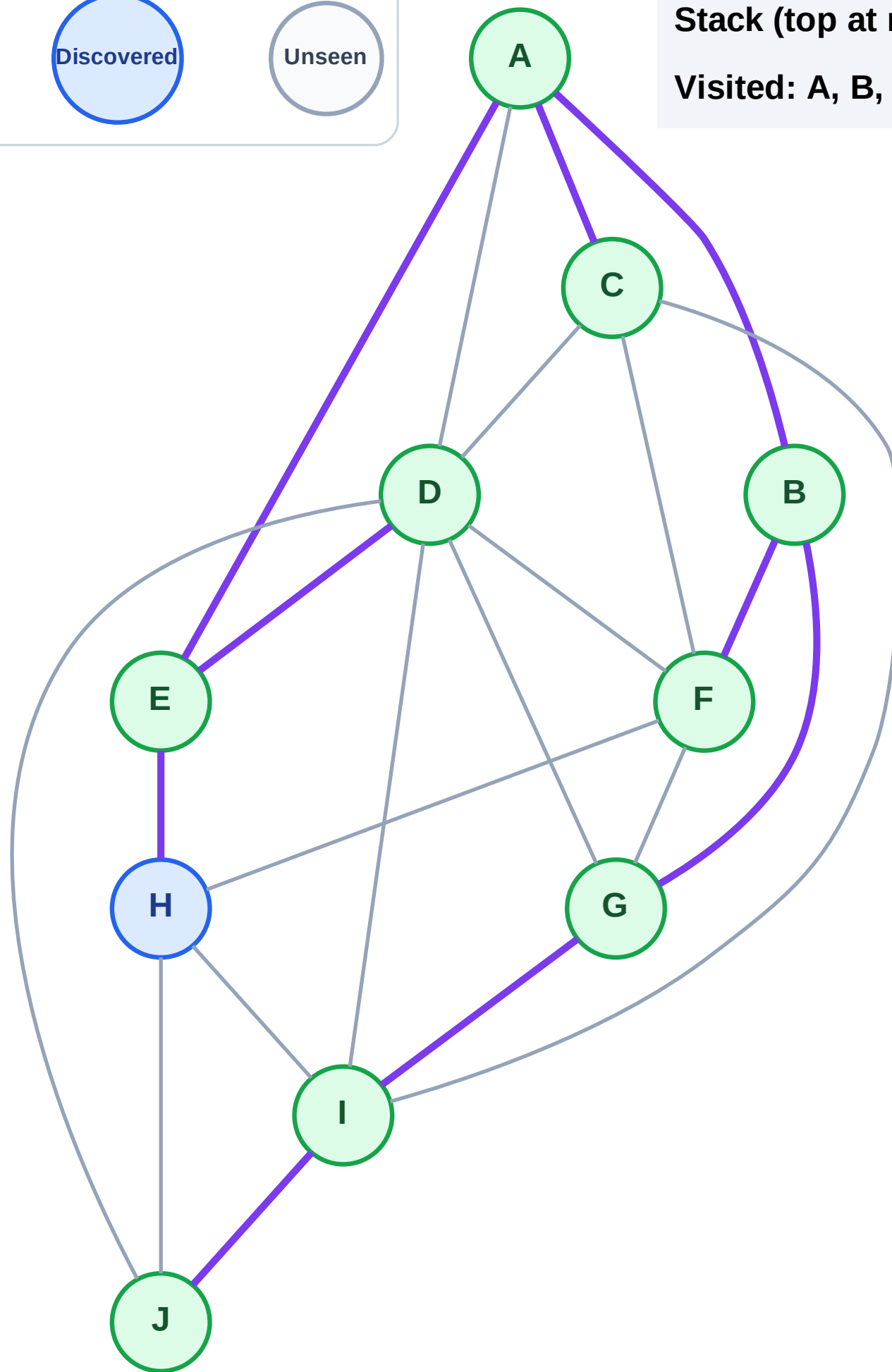
Step 19: No new neighbors from D

Legend



Stack (top at right): H

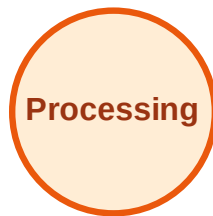
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

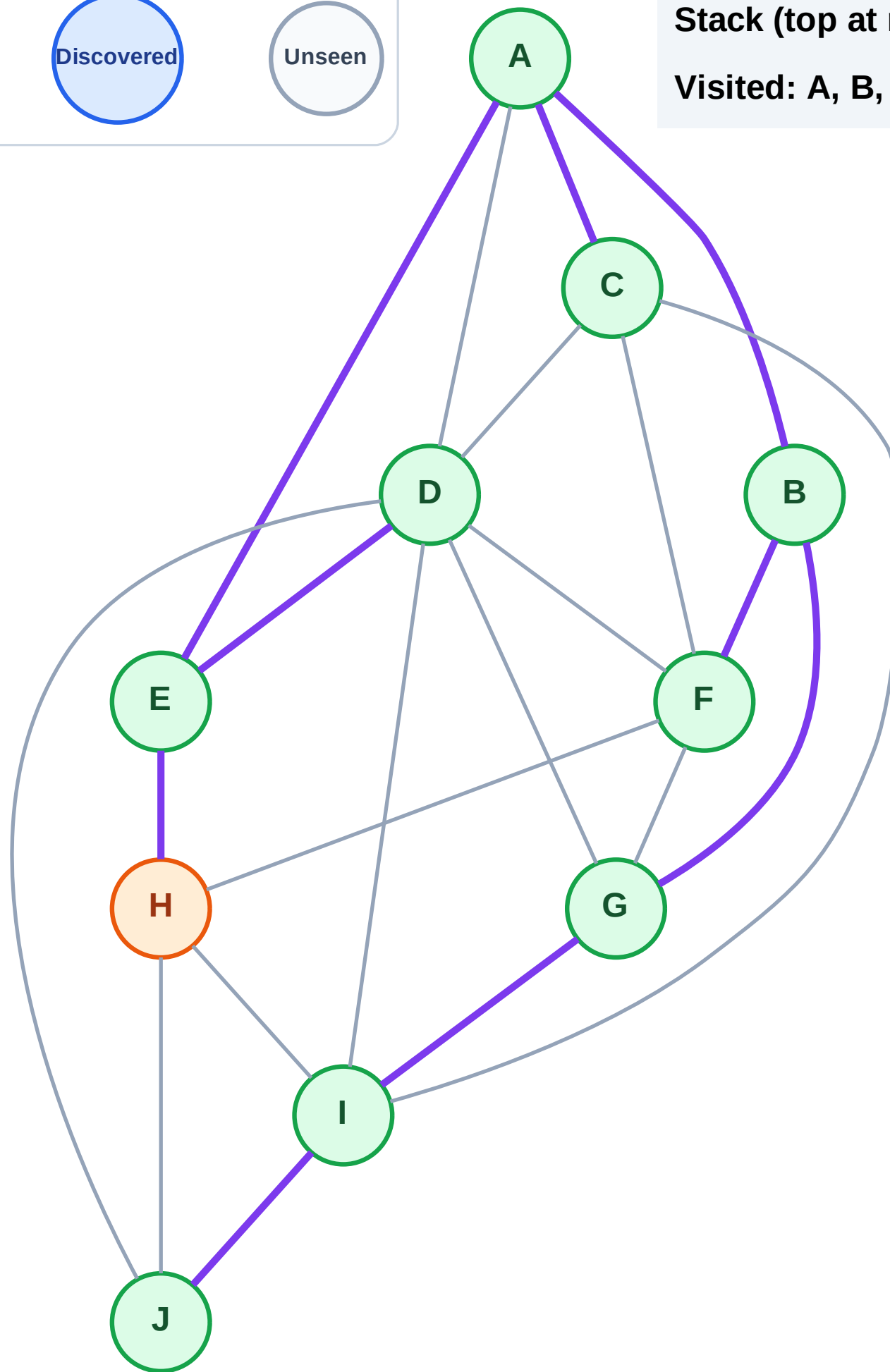
Step 20: Process node H

Legend



Stack (top at right): (empty)

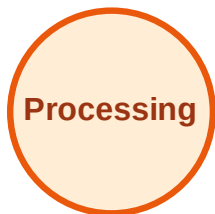
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

